OPTIMAL LINEAR FILTERING IN MEAN SQUARE SENSE: LINEAR PREDICTION

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Optimal Linear Filtering: Linear Prediction

- **Optimal linear filtering of a WSS random process**: it deals with the design of linear filters to process a class of signals with statistically similar characterstics. It is based on LMMSE estimation.
- It has many applications in various fields of engineering, such as processing of speech signals, image processing, noise suppression and power spectral estimation.
- The filter is designed to be **linear time-invariant (LTI)** and the optimality criterion is to minimize the **mean square error (MSE)** → **Orthogonality Principle**.
- The constraint that the filter should be LTI and that the criterion is to minimize the MSE implies that to design the optimal filter for pure linear prediction we need to know only the second order statistics of the process, i.e. the ACF or the PSD.

Optimal 1-Step Linear Prediction

- **Problem:** predict (i.e. estimate) the future value of a WSS random process $X[n]$, on the base of the observation of past values of the process itself (**pure prediction**) or on the base of noisy observations of the process (**filtering and prediction**).

- **Optimal linear predictor at 1-step of order *P* (pure prediction):** we want to estimate $X[n]$ given $X[n-1]$, $X[n-2]$, … , $X[n-P]$:

$$\hat{X}[n] = \sum_{k=1}^{P} h[k]X[n-k] = h[1]X[n-1] + \cdots + h[P]X[n-P]$$

- We could also express the predictor as follows:

$$\hat{X}[n+1] = \sum_{k=0}^{P-1} h[k]X[n-k] = h[0]X[n] + \cdots + h[P-1]X[n-P+1]$$

- In practice, it does not change anything. We will adopt the first choice.

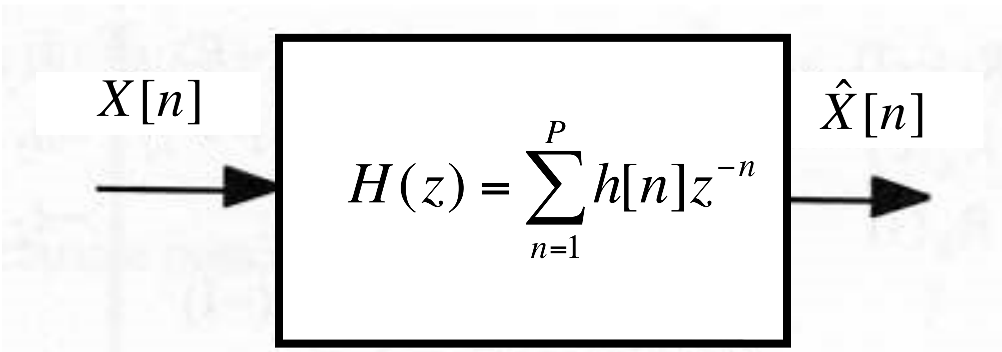
Optimal 1-Step Linear Prediction

- **Optimal 1-step linear predictor of order *P*:** we want to estimate $X[n]$ given the observed data $X[n-1]$, $X[n-2]$, … , $X[n-P]$:

$$\hat{X}[n] = \sum_{k=1}^{P} h[k]X[n-k] = h[1]X[n-1] + \cdots + h[P]X[n-P]$$

- $h[n]$ is the impulse response of the linear predictor: it is an FIR filter of order P at whose output we obtain an estimate of $X[n]$ for varying discrete-time n. Note that we are implicitly assuming that $h[0]=0$.

- The system function $H(z)$ of the linear predictor is obtained as the monolateral Z-transform of $h[n]$:



Optimal 1-Step Linear Prediction

- The **prediction error**, i.e. the estimation error, $\varepsilon[n]$ is given by:

$$\varepsilon[n] = X[n] - \hat{X}[n] = X[n] - \sum_{k=1}^{P} h[k] X[n-k]$$

- The prediction error $\varepsilon[n]$ can be thought as obtained by filtering the process $X[n]$ with an FIR filter having impulse response $e[n]$, called the **prediction error filter (PEF)**, given by:

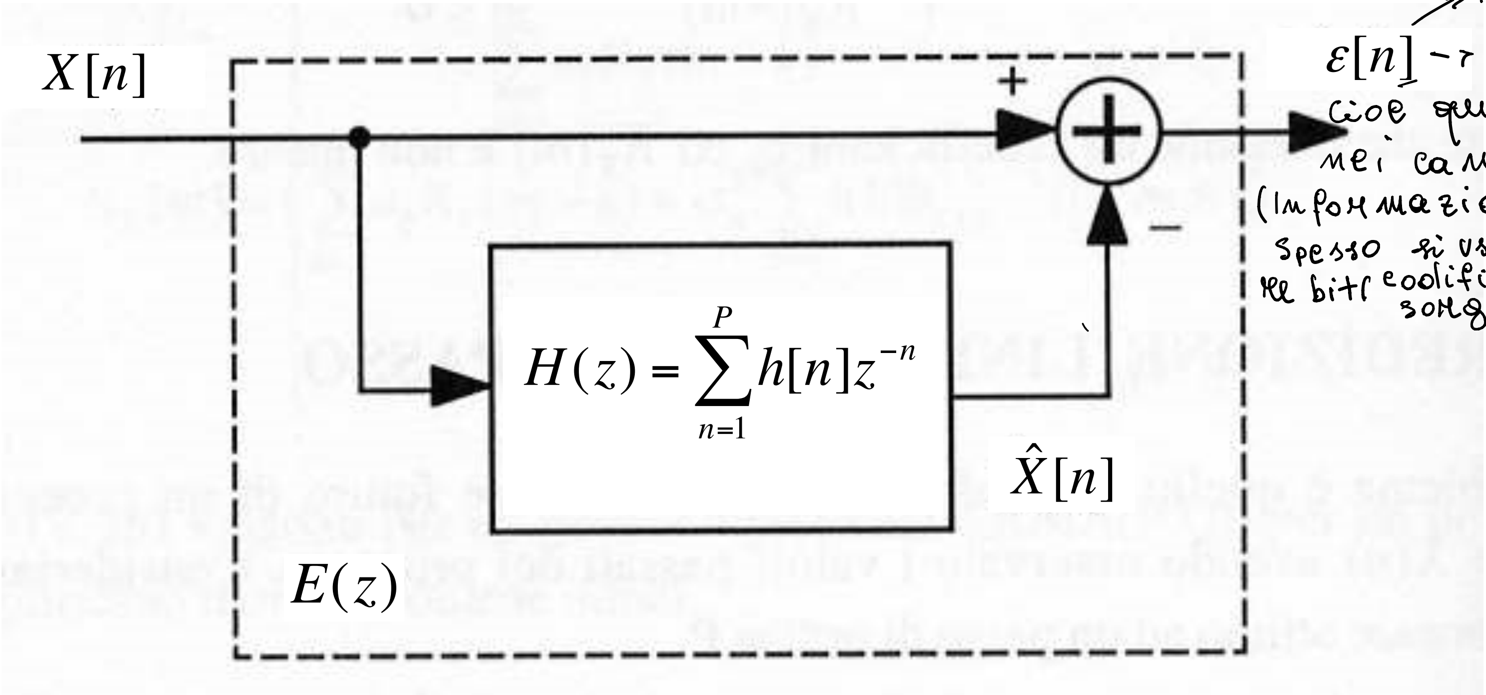
$$\varepsilon[n] = X[n] - \sum_{k=1}^{P} h[k] X[n-k] = e[n] \otimes X[n] \quad \Rightarrow \quad \varepsilon(z) = E(z) X(z)$$

Optimal 1-Step Linear Prediction

- The **prediction error filter** has system function *E*(*z*) given by:

$$\varepsilon(z) = E(z)X(z) = X(z) - H(z)X(z) = \left[1 - H(z)\right]X(z)$$

$$\Rightarrow \quad E(z) = 1 - H(z) = 1 - \sum_{n=1}^{P} h[n]z^{-n}$$



Parte innovativa

$\varepsilon[n]$ → la parte in più cioè quello non già cono= nei campioni usati.
(Informazione ridondante; spesso si usa l'errore per salva= re bit (codifica sorgente, infomed.)

Optimal 1-Step Linear Prediction

- The coefficients of the optimal linear predictor $h[n]$ are derived by minimizing the MSE, i.e. by applying the **Orthogonality Principle**:

↳ $\perp$ ai soli dati con cui si è stimato il valore.

$$\min_{\{h[n]\}} \sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} \quad \Rightarrow \quad E\left\{\varepsilon[n]X[n-m]\right\} = 0, \quad m = 1, 2, \cdots, P$$

$$\begin{aligned}
E\left\{\varepsilon[n]X[n-m]\right\} &= E\left\{\left(X[n] - \hat{X}[n]\right)X[n-m]\right\} \\
&= E\left\{\left(X[n] - \sum_{k=1}^{P} h[k]X[n-k]\right)X[n-m]\right\} \\
&= E\left\{X[n]X[n-m]\right\} - \sum_{k=1}^{P} h[k]E\left\{X[n-k]X[n-m]\right\} \\
&= R_X[m] - \sum_{k=1}^{P} h[k]R_X[m-k] = 0, \quad \underline{1 \le m \le P}
\end{aligned}$$

Optimal 1-Step Linear Prediction

$$R_X[m] = \sum_{k=1}^{P} h[k] R_X[m-k], \quad 1 \le m \le P$$

- This is typically called the discrete-time **Wiener-Hopf equation** for the **pure prediction problem**.
- The value of the MSE of the optimal linear predictor is also obtained by applying the **Orthogonality Principle** :

errore $\perp$ dati

$\hat{X} \rightarrow$ comb. lin errori $\rightarrow$ sono ortogonali

$$\sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{\varepsilon[n]\left(X[n] - \hat{X}[n]\right)\right\} = E\left\{\varepsilon[n]X[n]\right\}$$

$$= E\left\{\left(X[n] - \sum_{k=1}^{P} h[k]X[n-k]\right)X[n]\right\}$$

$$= R_X[0] - \sum_{k=1}^{P} h[k]R_X[k] \quad \Rightarrow \quad R_X[0] = \sum_{k=1}^{P} h[k]R_X[k] + \sigma_\varepsilon^2$$

Optimal 1-Step Linear Prediction

Augmente wiener-hopf

$$R_X[m] = \sum_{k=1}^{P} h[k] R_X[m-k] + \sigma_\varepsilon^2 \delta[m], \quad 0 \le m \le P$$

- It can be noted that the previous P+1 linear equations are equal to the P+1 linear equations obtained for the AR(P) processes, called **Yule-Walker equations**, that we recall here as:

$$R_X[m] = \sum_{k=1}^{P} a_k R_X[m-k] + \sigma_W^2 \delta[m], \quad m \ge 0$$

as long as we substitute a_k with $h[k]$ and the variance of $W[n]$ with the variance of $\varepsilon[n]$.

- This fact is even more evident if we recast the system of the P+1 linear equations of the predictor in matrix form.

Optimal 1-Step Linear Prediction

- The matrix form of the system of *P*+1 linear equations is:

$$\begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] & \cdots & R_X[-P] \\ R_X[1] & R_X[0] & \ddots & \ddots & R_X[-P+1] \\ R_X[2] & R_X[1] & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & R_X[-1] \\ R_X[P] & R_X[P-1] & \cdots & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} 1 \\ -h[1] \\ -h[2] \\ \vdots \\ -h[P] \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- and, by posing:

$$\mathbf{C}_X \triangleq \begin{bmatrix} R_X[0] & R_X[-1] & \cdots & R_X[-(P-1)] \\ R_X[1] & R_X[0] & \cdots & R_X[-(P-2)] \\ \vdots & \vdots & \ddots & \vdots \\ R_X[P-1] & R_X[P-2] & \cdots & R_X[0] \end{bmatrix}, \quad \mathbf{c}_X \triangleq \begin{bmatrix} R_X[1] \\ R_X[2] \\ \vdots \\ R_X[P] \end{bmatrix}, \quad \mathbf{h} \triangleq \begin{bmatrix} h[1] \\ h[2] \\ \vdots \\ h[P] \end{bmatrix}$$

Optimal 1-Step Linear Prediction

$$\begin{bmatrix} R_X[0] & \mathbf{c}_X^T \\ \mathbf{c}_X & \mathbf{C}_X \end{bmatrix} \begin{bmatrix} 1 \\ -\mathbf{h} \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 \\ \mathbf{0} \end{bmatrix} \Rightarrow \begin{cases} R_X[0] - \mathbf{c}_X^T \mathbf{h} = \sigma_\varepsilon^2 \\ \mathbf{c}_X - \mathbf{C}_X \mathbf{h} = \mathbf{0} \end{cases}$$

- We solve this matrix equation with respect to **h** by using the **Levinson-Durbin** or the **Schur** algorithms. If the symmetric and Toeplitz covariance matrix $\mathbf{C}_X$ is full rank, we have:

$$\begin{cases} \mathbf{h} = \mathbf{C}_X^{-1} \mathbf{c}_X \\ \sigma_\varepsilon^2 = R_X[0] - \mathbf{c}_X^T \mathbf{C}_X^{-1} \mathbf{c}_X \end{cases}$$

- This result tells us that, if $X[n]$ is an AR(P) process, since **h** and **a** satisfy the same equations, then:

Non si può andare sotto questo errore (se il processo è regolare)

$$\begin{cases} \mathbf{h} = \mathbf{a} \\ \sigma_\varepsilon^2 = \sigma_W^2 \end{cases}$$

Optimal 1-Step Linear Prediction

- Hence, if *X*[*n*] is an AR(*P*) process:

line regressione dei dati usati

$$X[n] = \sum_{k=1}^{P} a_k X[n-k] + W[n], \quad \sigma_W^2 = E\{W^2[n]\}$$

Parte non stimabile (innovativa)

$$W[n] \equiv \text{ innovation process of } X[n]$$

- then $h[k]=a_k$ for $k=1,2,\ldots,P$ and 0 otherwise. Finally:

$$\hat{X}[n] = \sum_{k=1}^{P} a_k X[n-k] = a_1 X[n-1] + \cdots + a_P X[n-P]$$

$$\varepsilon[n] = X[n] - \hat{X}[n] = X[n] - \sum_{k=1}^{P} a_k X[n-k] = W[n]$$

$$X[n] \longrightarrow \boxed{\Gamma(z)} \longrightarrow W[n]$$

$\updownarrow$ sono =

$$\longrightarrow \boxed{E(z)} \longrightarrow \varepsilon(n)$$

$$E(z) = \Gamma(z)$$

$$H(z) = 1 - \Gamma(z) = 1 - \frac{1}{L(z)}$$

prediction error = innovation process of $X[n]$

↳ Dati una volta tolta la ridondanza.

$$\sigma_\varepsilon^2 = E\{\varepsilon^2[n]\} = E\{W^2[n]\} = \sigma_W^2$$

Optimal 1-Step Linear Prediction

- The same result can be obtained by using a different approach → The MMSE estimator of $X[n]$ given the observations $X[n-1]$, $X[n-2]$, $X[n-3]$, … , is the conditional mean of $X[n]$ given the observations:

$$\hat{X}_{MMSE}[n] = E\left\{X[n] \middle| X[n-1], X[n-2], X[n-3], \cdots\right\}$$

$$= E\left\{\sum_{k=1}^{P} a_k X[n-k] + W[n] \middle| X[n-1], X[n-2], X[n-3], \cdots\right\}$$

$$= \sum_{k=1}^{P} a_k X[n-k] + E\left\{W[n] \middle| X[n-1], X[n-2], X[n-3], \cdots\right\}$$

$$= \sum_{k=1}^{P} a_k X[n-k] + E\left\{W[n]\right\}$$

$$= \sum_{k=1}^{P} a_k X[n-k]$$

If $W[n]$ is independent of the observations $X[n-1]$, $X[n-2]$, $X[n-3]$, … i.e. if the $\{W[n]\}$'s are IID.

Optimal 1-Step Linear Prediction

- Then, for an **AR(P) process**:

$$X[n] = \sum_{k=1}^{P} a_k X[n-k] + W[n], \quad \sigma_W^2 = E\left\{W^2[n]\right\}$$

$$W[n] \equiv \text{innovation process of } X[n]$$

the optimal (in the MSE sense) **linear predictor** is a linear time-invariant (LTI) finite impulse response (FIR) filter of order P with impulse response **h**=**a**:

$$\hat{X}[n] = \sum_{k=1}^{P} a_k X[n-k], \qquad \sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{W^2[n]\right\} = \sigma_W^2$$

- It can be noted that if $X[n]$ is an AR(P) process, the observations $X[n-P-1]$, $X[n-P-2]$, … , are not useful for the estimation of $X[n]$.

Optimal 1-Step Linear Prediction

Example: *X*[*n*] is an AR(*2*) process, $X[n] = a_1 X[n-1] + a_2 X[n-2] + W[n]$

il numero di campioni p dipendono dalla corr. (maggiore t. corr. → maggior numero di val.)

Proc. banda stretta tcorr lungo → tanti campioni

Proc. banda larga ..... opposto

- The PSD can be immediately derived:

$$S_X\left(e^{j2\pi f}\right) = \sigma_W^2 \left|L\left(e^{j2\pi f}\right)\right|^2 = \frac{\sigma_W^2}{\left|1 - a_1 e^{-j2\pi f} - a_2 e^{-j4\pi f}\right|^2} = \frac{\sigma_W^2}{\left|e^{j2\pi f} - p_1\right|^2 \left|e^{j2\pi f} - p_2\right|^2}$$

- The PSD of an AR(2) process with complex conjugate poles is pass-band, with central frequency f_0 and bandwidth *B* which depends on *d*: the closer is *d* to 1 and the smaller is *B*.

$$d \triangleq |p_1| = |p_2|$$

$$\underline{h} = \begin{bmatrix} h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ 0 \end{bmatrix}$$

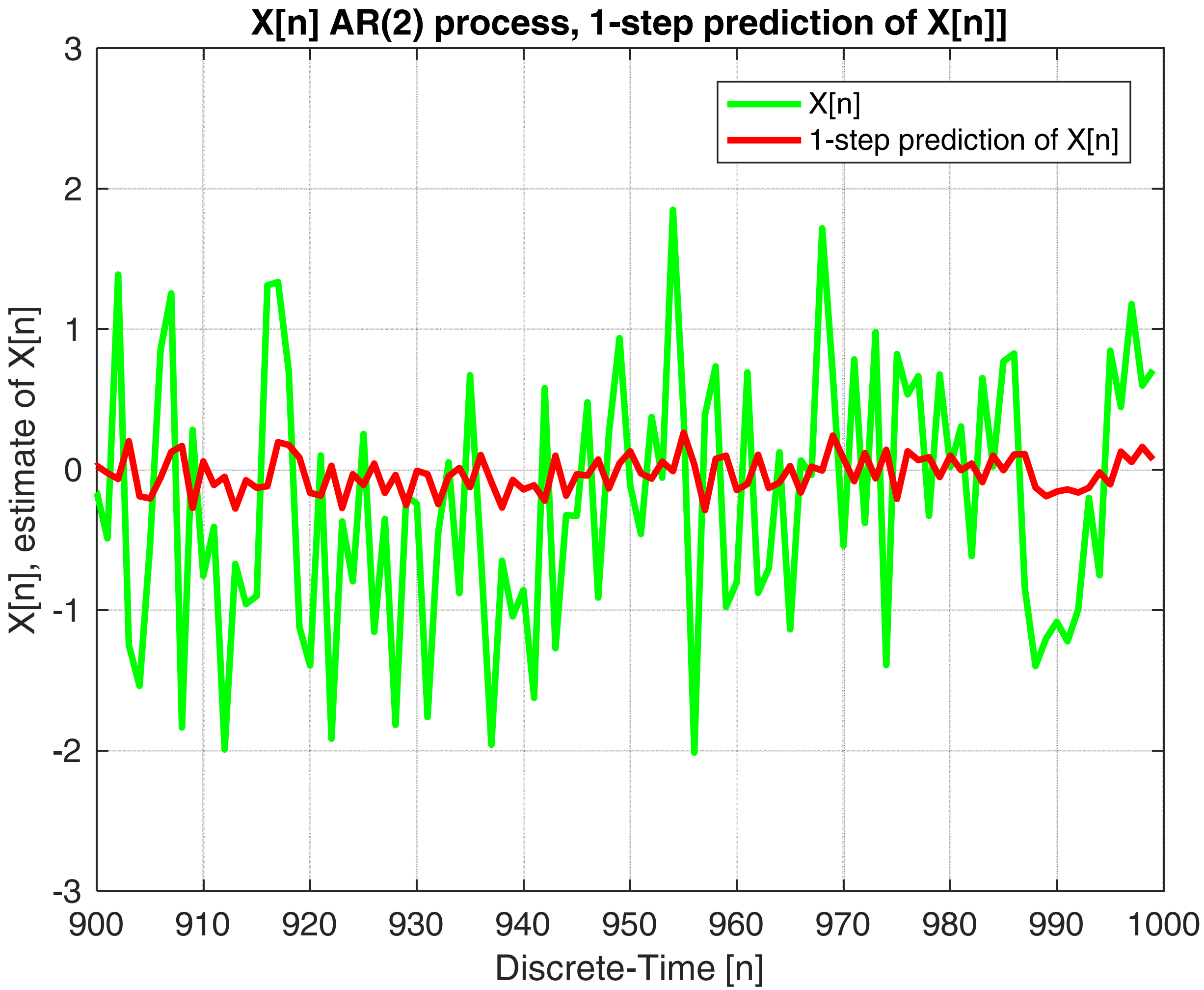
- The optimal linear predictor is a LTI FIR filter of order 2:

$$\hat{X}[n] = a_1 X[n-1] + a_2 X[n-2]$$

Se ho un processo di ordine p, non ha senso cercare un predittore di ordine p+1

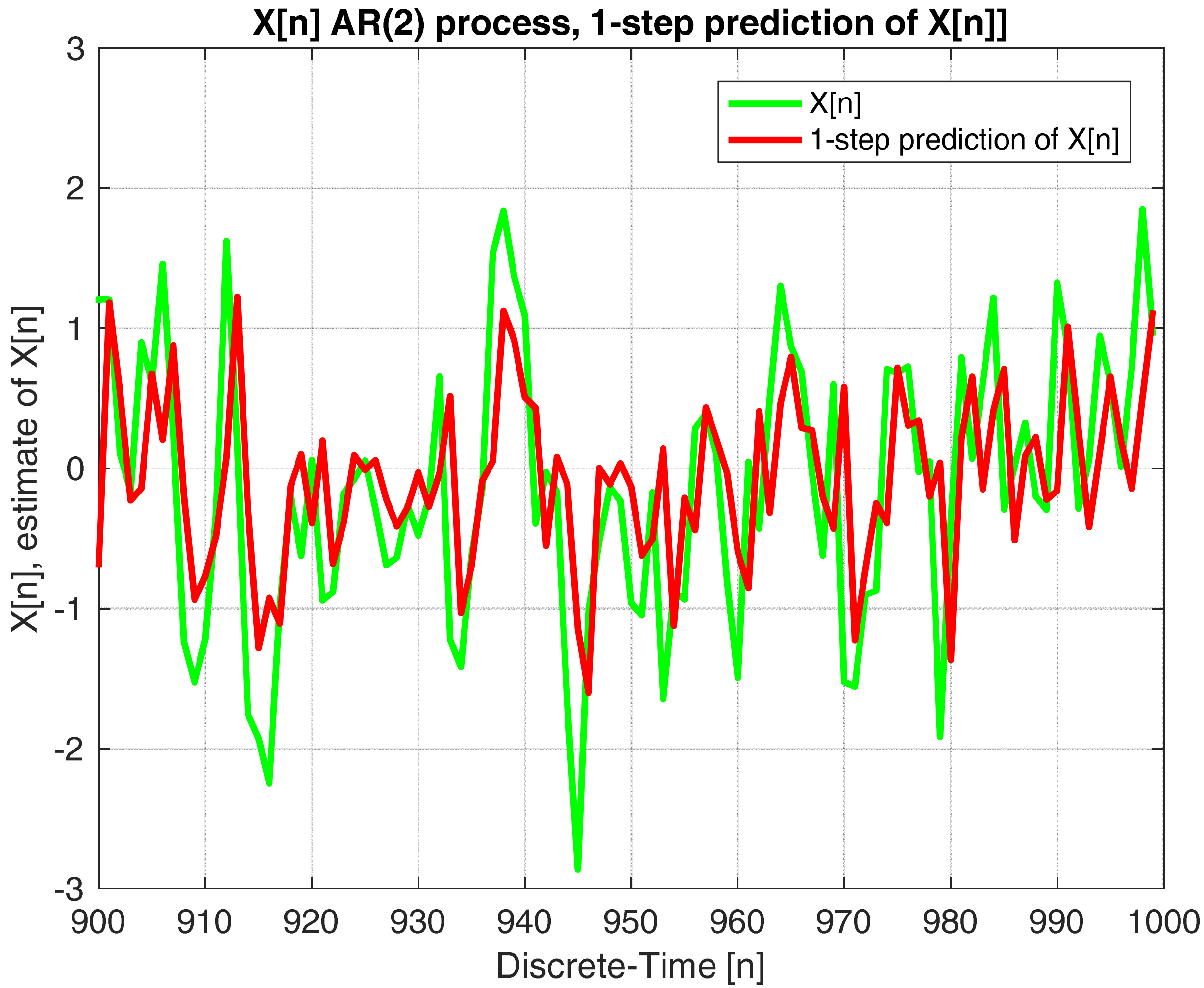
Optimal 1-Step Linear Prediction

$$\sigma_X^2 = 1, \quad d = 0.1, \quad f_0 = 1/8$$



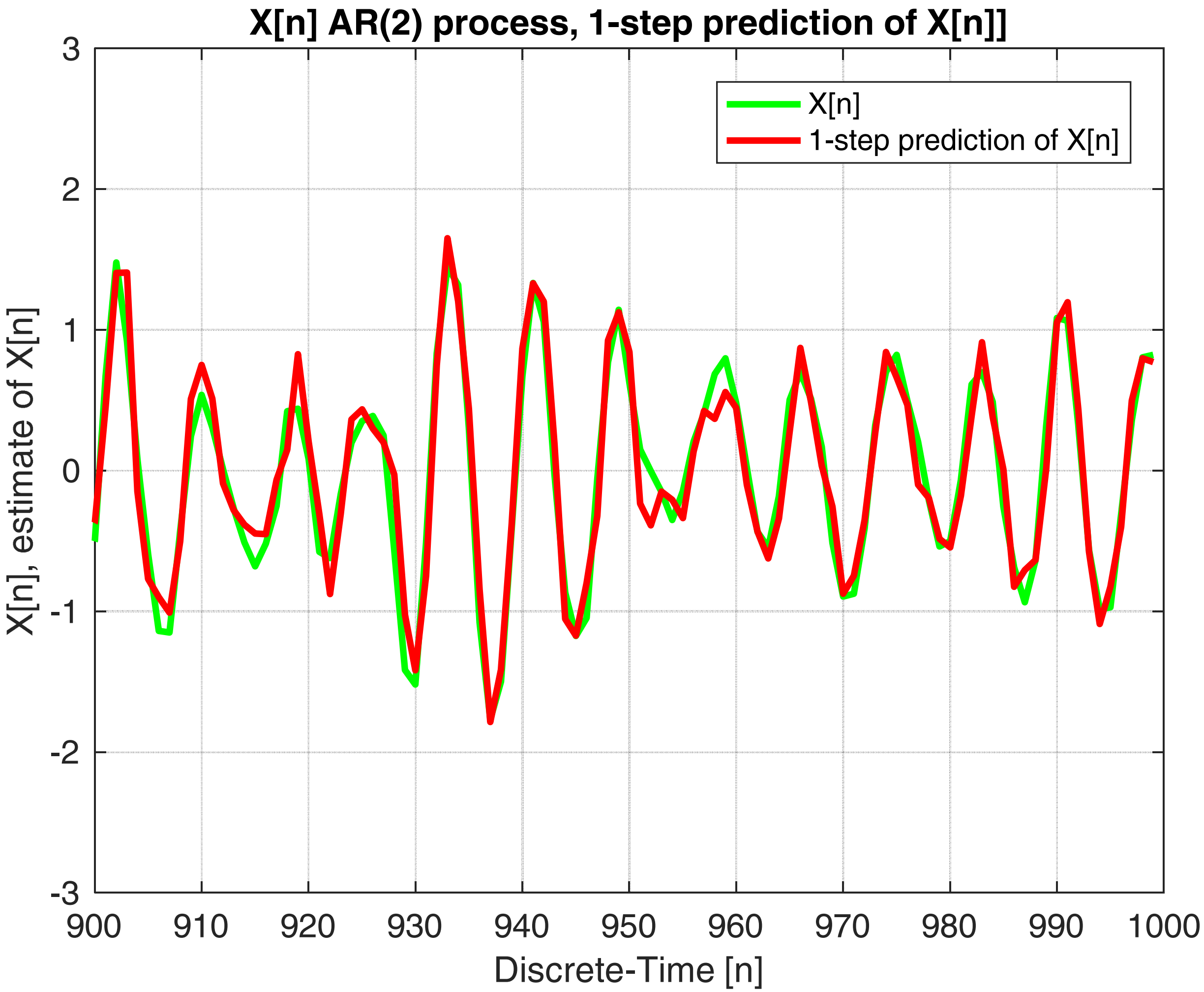
Optimal 1-Step Linear Prediction

$$\sigma_X^2 = 1, \quad d = 0.5, \quad f_0 = 1/8$$



Optimal 1-Step Linear Prediction

$$\sigma_X^2 = 1, \quad d = 0.98, \quad f_0 = 1/8$$



Optimal 1-Step Linear Prediction

$$\hat{X}[n] = \sum_{k=1}^{2} a_k X[n-k], \qquad MSE\left\{\hat{X}[n]\right\} = \sigma_W^2$$

- These numerical results have been obtained by keeping constant the power of the process *X*[*n*] and by varying the modulo *d* of the two poles.
- The MSE of the **optimal 1-step linear predictor** is equal to the power of the innovation process, i.e.

$$MSE\left\{\hat{X}[n]\right\} = E\left\{\left(X[n] - \hat{X}[n]\right)^2\right\} = \sigma_W^2 \equiv \sigma_W^2(\mathbf{p}, \sigma_X^2)$$

- It can be noted from the previous plots, that the power of *W*[*n*] decreases with *d* (if the power of *X*[*n*] is kept constant), with a consequent decrease of the MSE.

Optimal *L*-Steps Linear Prediction

- ***L*-steps optimal linear predictor of order *P*:** our aim is to estimate $X[n+L-1]$ given $X[n-1]$, $X[n-2]$, … , $X[n-P]$:

$$\hat{X}[n+L-1]=\sum_{k=1}^{P}h[k]X[n-k]=h[1]X[n-1]+\cdots+h[P]X[n-P]$$

- As before, the coefficients of the prediction filter $h[n]$ are obtained by minimizing the MSE, i.e. by applying the **Orthogonality Principle**:

$$\min_{\{h[n]\}}\sigma_\varepsilon^2=E\left\{\varepsilon^2[n+L-1]\right\}\quad\Rightarrow\quad E\left\{\varepsilon[n+L-1]X[n-m]\right\}=0,\quad 1\le m\le P$$

$$\varepsilon[n+L-1]=X[n+L-1]-\hat{X}[n+L-1]=X[n+L-1]-\sum_{k=1}^{P}h[k]X[n-k]$$

Optimal *L*-Steps Linear Prediction

- **Orthogonality Principle**:

$$E\left\{\varepsilon[n+L-1]X[n-m]\right\}=E\left\{\left(X[n+L-1]-\sum_{k=1}^{P}h[k]X[n-k]\right)X[n-m]\right\}$$

$$=E\left\{X[n+L-1]X[n-m]\right\}-\sum_{k=1}^{P}h[k]E\left\{X[n-k]X[n-m]\right\}$$

$$=R_X[m+L-1]-\sum_{k=1}^{P}h[k]R_X[m-k]=0,\quad 1\le m\le P$$

$$R_X[m+L-1]=\sum_{k=1}^{P}h[k]R_X[m-k],\quad 1\le m\le P$$

- If L=1, we go back to the 1-step linear prediction problem previously discussed.

Optimal *L*-Steps Linear Prediction

- By applying the **Orthogonality Principle**, we can obtain the MSE of the optimal linear predictor:

$$\begin{aligned}
\sigma_\varepsilon^2 &= E\left\{\varepsilon^2[n+L-1]\right\} = E\left\{\varepsilon[n+L-1]\left(X[n+L-1]-\hat{X}[n+L-1]\right)\right\} \\
&= E\left\{\varepsilon[n+L-1]X[n+L-1]\right\} \\
&= E\left\{\left(X[n+L-1]-\sum_{k=1}^{P}h[k]X[n-k]\right)X(n+L-1)\right\} \\
&= R_X[0]-\sum_{k=1}^{P}h[k]R_X[k+L-1]
\end{aligned}$$

- As before, the *P*+1 linear equations can be rewritten in matrix form as follows.

Optimal *L*-Steps Linear Prediction

- The system of P+1 linear equations becomes:

$$\begin{bmatrix} R_X[0] & R_X[-L] & R_X[-(L+1)] & \cdots & R_X[-(L-1+P)] \\ R_X[L] & R_X[0] & \ddots & \ddots & R_X[-(P-1)] \\ R_X[L+1] & R_X[1] & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & R_X[-1] \\ R_X[L-1+P] & R_X[P-1] & \cdots & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} 1 \\ -h[1] \\ -h[2] \\ \vdots \\ -h[P] \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- and, by setting:

$$\mathbf{C}_X \triangleq \begin{bmatrix} R_X[0] & R_X[-1] & \cdots & R_X[-(P-1)] \\ R_X[1] & R_X[0] & \cdots & R_X[-(P-2)] \\ \vdots & \vdots & \ddots & \vdots \\ R_X[P-1] & R_X[P-2] & \cdots & R_X[0] \end{bmatrix}, \quad \mathbf{c}_X \triangleq \begin{bmatrix} R_X[L] \\ R_X[L+1] \\ \vdots \\ R_X[L+P-1] \end{bmatrix}, \quad \mathbf{h} \triangleq \begin{bmatrix} h[1] \\ h[2] \\ \vdots \\ h[P] \end{bmatrix}$$

Optimal *L*-Steps Linear Prediction

- The linear system has the same structure regardless the value of *L*, what changes is the vector $\mathbf{c}_X$:

$$\begin{bmatrix} R_X[0] & \mathbf{c}_X^T \\ \mathbf{c}_X & \mathbf{C}_X \end{bmatrix} \begin{bmatrix} 1 \\ -\mathbf{h} \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 \\ \mathbf{0} \end{bmatrix} \Rightarrow \begin{cases} R_X[0] - \mathbf{c}_X^T \mathbf{h} = \sigma_\varepsilon^2 \\ \mathbf{c}_X - \mathbf{C}_X \mathbf{h} = \mathbf{0} \end{cases}$$

- At this point, we can solve the linear matrix equation w.r.t. **h** by using the **Levinson-Durbin** or the **Schur** algorithm. If the covariance matrix $\mathbf{C}_X$ is full rank, we obtain:

$$\begin{cases} \mathbf{h} = \mathbf{C}_X^{-1} \mathbf{c}_X \\ \sigma_\varepsilon^2 = R_X[0] - \mathbf{c}_X^T \mathbf{C}_X^{-1} \mathbf{c}_X \end{cases}$$

Optimal *L*-Steps Linear Prediction: AR(1) Process

- ***Example***: consider an AR(1) process and derive the optimal **1-step** and ***L*-steps linear predictor** of order *P*.

$$X[n] = \rho X[n-1] + W[n] \quad (a_1 \equiv \rho)$$

↳ coeff. corr. ad un passo

- **Optimal 1-step linear predictor of order *P*** for an **AR(1)** process.
  The 1-step optimal linear predictor and its MSE are given by:

$$\begin{cases} \hat{X}[n] = \rho X[n-1] \\ \sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{W^2[n]\right\} = \sigma_W^2 = \sigma_X^2(1-\rho^2) \end{cases}$$

Optimal *L*-Steps Linear Prediction: AR(1) Process

- **Optimal *L*-steps linear predictor of order *P*** for an **AR(1)** process.

$$R_X[m] = \sigma_X^2 \rho^{|m|} = \frac{\sigma_W^2}{1-\rho^2}\rho^{|m|}$$

- As previously discussed, the structure of the linear system is invariant w.r.t. *L*; what is different is only the vector $\mathbf{c}_X$:

$$\mathbf{c}_X \triangleq \begin{bmatrix} R_X[L] \\ R_X[L+1] \\ \vdots \\ R_X[L+P-1] \end{bmatrix} = \begin{bmatrix} \sigma_X^2 \rho^{|L|} \\ \sigma_X^2 \rho^{|L+1|} \\ \vdots \\ \sigma_X^2 \rho^{|L+P-1|} \end{bmatrix} = \rho^{|L-1|} \begin{bmatrix} \sigma_X^2 \rho \\ \sigma_X^2 \rho^2 \\ \vdots \\ \sigma_X^2 \rho^{|P|} \end{bmatrix} = \rho^{|L-1|}\mathbf{c}_X\left(L=1\right)$$

Optimal L-Steps Linear Prediction: AR(1) Process

risp. impulsiva predittore ad un passo

$$\mathbf{h} = \mathbf{C}_X^{-1}\mathbf{c}_X = \rho^{L-1}\mathbf{C}_X^{-1}\mathbf{c}_X(L=1) = \rho^{L-1}\mathbf{h}(L=1) \quad \Rightarrow \quad h[k] = \begin{cases} \rho^L, & k=1 \\ 0, & otherwise \end{cases}$$

$$\hat{X}[n+L-1] = \rho^L X[n-1]$$

$$\sigma_\varepsilon^2 = R_X[0] - \sum_{k=1}^{P} h[k]R_X[k+L-1] = R_X[0] - h[1]R_X[L]$$

$$= \sigma_X^2 - \rho^L \sigma_X^2 \rho^L = \sigma_X^2\left(1-\rho^{2L}\right)$$

- Hence, the MSE increases by increasing L, so the estimation performance gets worse. In fact:

$$\sigma_\varepsilon^2 = \sigma_X^2\left(1-\rho^{2L}\right) > \sigma_X^2\left(1-\rho^2\right) \quad \forall L > 1$$

↳ più avanti osservo più la correlazione vale poco.

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- ***Example***: Optimal 1-step linear predictor of a parametric sinusoidal process:

$$X[n] = A\cos\left[2\pi f_0 n + \theta\right]$$

- As previously discussed, $X[n]$ has a discrete PSD, so it is a **predictable process**, not a regular process.

spettro a righe
→ comp. periodiche
→ corr. periodica

- **Assumptions**: the amplitude A is modelled as a Rayleigh r.v., whereas the phase θ is modelled as a r.v. uniformly distributed in $[-\pi,\pi)$. The amplitude and the phase are mutually independent.

$$A \in \mathcal{R}(\sigma_A^2) \qquad \theta \in \mathcal{U}(-\pi,\pi) \qquad A \text{ and } \theta \text{ independent}$$

$$f_A(a) = \frac{a}{\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a), \quad E\{A^2\} = 2\sigma_A^2$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- **Problem**: find the optimal 1-step linear predictor of order P=3.

$$\hat{X}[n] = \sum_{k=1}^{3} h[k]X[n-k] = h[1]X[n-1] + h[2]X[n-2] + h[3]X[n-3]$$

- The impulse response of the optimal 1-step linear predictor of order 3, a FIR(3) filter, is obtained by solving the **Wiener-Hopf equation**, which in matrix form is given by:

$$\begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] & R_X[-3] \\ R_X[1] & R_X[0] & R_X[-1] & R_X[-2] \\ R_X[2] & R_X[1] & R_X[0] & R_X[-1] \\ R_X[3] & R_X[2] & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} 1 \\ -h[1] \\ -h[2] \\ -h[3] \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- By setting, as usual:

$$\mathbf{C}_X \triangleq \begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] \\ R_X[1] & R_X[0] & R_X[-1] \\ R_X[2] & R_X[1] & R_X[0] \end{bmatrix}, \quad \mathbf{c}_X \triangleq \begin{bmatrix} R_X[1] \\ R_X[2] \\ R_X[3] \end{bmatrix}, \quad \mathbf{h} \triangleq \begin{bmatrix} h[1] \\ h[2] \\ h[3] \end{bmatrix}$$

- we obtain:

$$\begin{bmatrix} R_X[0] & \mathbf{c}_X^T \\ \mathbf{c}_X & \mathbf{C}_X \end{bmatrix} \begin{bmatrix} 1 \\ -\mathbf{h} \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 \\ \mathbf{0} \end{bmatrix} \Rightarrow \begin{cases} \sigma_\varepsilon^2 = R_X[0] - \mathbf{c}_X^T \mathbf{h} \\ \mathbf{C}_X \mathbf{h} = \mathbf{c}_X \end{cases}$$

$$\text{if } \det\{\mathbf{C}_X\} \neq 0$$

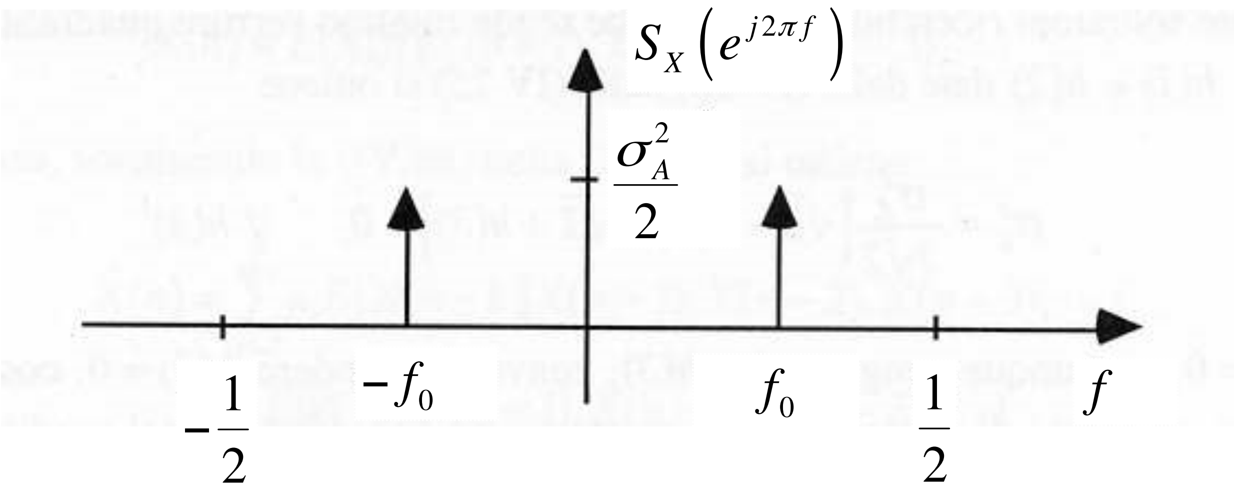
$$\begin{cases} \mathbf{h} = \mathbf{C}_X^{-1} \mathbf{c}_X \\ \sigma_\varepsilon^2 = R_X[0] - \mathbf{c}_X^T \mathbf{C}_X^{-1} \mathbf{c}_X \end{cases}$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- For this particular case, we have:

$$R_X[m] = \sigma_A^2 \cos\left[2\pi f_0 m\right]$$

$$S_X\left(e^{j2\pi f}\right) = 2\pi \cdot \frac{\sigma_A^2}{2}\left[\delta\left(e^{j2\pi f} - e^{j2\pi f_0}\right) + \delta\left(e^{j2\pi f} - e^{-j2\pi f_0}\right)\right]$$



- Discrete PSD → the process is predictable (not regular).

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- As we will see, <u>for predictable processes, it is possible to find a linear predictor of a suitable order *P* such that its MSE is zero</u>.

- By substituting the ACF values in the **Wiener-Hopf equation**:

$$
\mathbf{C}_X \mathbf{h} = \mathbf{c}_X \quad \Rightarrow \begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] \\ R_X[1] & R_X[0] & R_X[-1] \\ R_X[2] & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} R_X[1] \\ R_X[2] \\ R_X[3] \end{bmatrix}
$$

$$
\sigma_A^2 \begin{bmatrix} 1 & \cos[2\pi f_0] & \cos[4\pi f_0] \\ \cos[2\pi f_0] & 1 & \cos[2\pi f_0] \\ \cos[4\pi f_0] & \cos[2\pi f_0] & 1 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \\ h[3] \end{bmatrix} = \sigma_A^2 \begin{bmatrix} \cos[2\pi f_0] \\ \cos[4\pi f_0] \\ \cos[6\pi f_0] \end{bmatrix}
$$

Il rango é ≤ ad il numero di Righe

→ rank$(C_x) \leq L$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

$$\begin{bmatrix} 1 & \cos[2\pi f_0] & \cos[4\pi f_0] \\ \cos[2\pi f_0] & 1 & \cos[2\pi f_0] \\ \cos[4\pi f_0] & \cos[2\pi f_0] & 1 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} \cos[2\pi f_0] \\ \cos[4\pi f_0] \\ \cos[6\pi f_0] \end{bmatrix}$$

- The problem is that $\mathbf{C}_X$ is not invertible, in fact:

$$\det\{\mathbf{C}_X\} = 0, \quad rank\{\mathbf{C}_X\} = 2$$

- The system has ∞^1 solutions, then we can derive $h[1]$ and $h[2]$ as a function of $h[3]$. To this purpose, let us consider the first two equations, that are linearly independent:

$$\begin{bmatrix} 1 & \cos[2\pi f_0] & \cos[4\pi f_0] \\ \cos[2\pi f_0] & 1 & \cos[2\pi f_0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} \cos[2\pi f_0] \\ \cos[4\pi f_0] \end{bmatrix}$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- Then, we can move the term that contains *h*[3] to the right-hand term, so we obtain the 2x2 linear system that can be solved as usual:

$$\begin{bmatrix} 1 & \cos[2\pi f_0] \\ \cos[2\pi f_0] & 1 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} \cos[2\pi f_0] - \cos[4\pi f_0]h[3] \\ \cos[4\pi f_0] - \cos[2\pi f_0]h[3] \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 1 & \cos[2\pi f_0] \\ \cos[2\pi f_0] & 1 \end{bmatrix}^{-1} \begin{bmatrix} \cos[2\pi f_0] - \cos[4\pi f_0]h[3] \\ \cos[4\pi f_0] - \cos[2\pi f_0]h[3] \end{bmatrix}$$

→ in base al valore che inserisco ho soluzione → infinite soluzioni poiché sono tutti ottimi → errore più picc. possibile, quindi discrimino rispetto h[3] il meno complesso possibile h[3]=0, FIR di ordine 2.

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{1}{\sin^2[2\pi f_0]} \begin{bmatrix} 1 & -\cos[2\pi f_0] \\ -\cos[2\pi f_0] & 1 \end{bmatrix} \begin{bmatrix} \cos[2\pi f_0] - \cos[4\pi f_0]h[3] \\ \cos[4\pi f_0] - \cos[2\pi f_0]h[3] \end{bmatrix}$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- For each value of $h[3]$ we obtain a different solution. However, it can be shown that all the ∞^1 solutions have the same MSE. Consequently, we can select the value of $h[3]$ that minimizes the number of data to be saved in a memory and processed $\rightarrow$ $h[3]=0$.

- In this case, the optimal 1-step linear predictor (of order P=2) is:

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{1}{\sin^2[2\pi f_0]} \begin{bmatrix} 1 & -\cos[2\pi f_0] \\ -\cos[2\pi f_0] & 1 \end{bmatrix} \begin{bmatrix} \cos[2\pi f_0] \\ \cos[4\pi f_0] \end{bmatrix}$$

$$= \frac{1}{\sin^2[2\pi f_0]} \begin{bmatrix} \cos[2\pi f_0]\left(1-\cos[4\pi f_0]\right) \\ \cos[4\pi f_0]-\cos^2[2\pi f_0] \end{bmatrix}$$

$$= \frac{1}{\sin^2[2\pi f_0]} \begin{bmatrix} 2\cos[2\pi f_0]\sin^2[2\pi f_0] \\ \cos^2[2\pi f_0]-1 \end{bmatrix}$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 2\cos[2\pi f_0] \\ -1 \end{bmatrix}$$

$$\Downarrow$$

$$\hat{X}[n] = 2\cos[2\pi f_0] X[n-1] - X[n-2]$$

- Note that f_0 needs to be a-priori known, that is the ACF/PSD of $X[n]$ has to be known.

Sono predicibili poiché arrivo ad un errore nullo.

- The MSE of the optimal predictor is:

$$\sigma_\varepsilon^2 = R_X[0] - \sum_{k=1}^{3} h[k] R_X[k] = \sigma_A^2 \left[1 - h[1]\cos[2\pi f_0] - h[2]\cos[4\pi f_0]\right]$$

$$= \sigma_A^2 \left[1 - 2\cos^2[2\pi f_0] + \cos[4\pi f_0]\right] = 0 \quad \forall f_0$$

$$\varepsilon[n] = 0 \quad \Rightarrow \quad X[n] = \hat{X}[n] = 2\cos[2\pi f_0] X[n-1] - X[n-2]$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- We proved that, for the parametric process at hand, that is characterized by a discrete PSD, the prediction error is zero. In this case, we get an *MSE*=0 with a linear predictor of order *P*=2. Note that 2 is also the number of the discrete lines in the PSD of $X[n]$.
- In general, the lowest predictor order *P* that guarantees *MSE*=0 is equal to the number *L* of discrete lines in the PSD.
- *MSE*=0 implies that the samples of $X[n]$ satisfy a linear difference equation, with constant coefficients similar to the one that characterizes an AR(2) without $W[n]$, or in other words, the sinusoidal process can be interpreted as a degenerate AR(2) process, i.e. an AR(2) process whose innovation process has a zero power:

Degenere

$$X[n] = 2\cos\left[2\pi f_0\right] X[n-1] - X[n-2]$$ --> poli sul cerchio

oscillatore

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- The characteristic equation is given by:

$$z^2 - 2\cos\left[2\pi f_0\right] z + 1 = 0$$

- whose solutions are:

$$p_{1,2} = \frac{2\cos\left[2\pi f_0\right] \pm \sqrt{4\cos^2\left[2\pi f_0\right] - 4}}{2}$$

$$= \cos\left[2\pi f_0\right] \pm \sqrt{\cos^2\left[2\pi f_0\right] - 1}$$

$$= \cos\left[2\pi f_0\right] \pm \sqrt{-\sin^2\left[2\pi f_0\right]}$$

$$= \cos\left[2\pi f_0\right] \pm j\left|\sin\left[2\pi f_0\right]\right|$$

Optimal 1-Step Linear Prediction: Cosinusoidal Process

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- The poles are complex and conjugated values and belong to the unit circle:

$$p_{1,2} = \cos\left[2\pi f_0\right] \pm j\left|\sin\left[2\pi f_0\right]\right| = \begin{cases} e^{\pm j2\pi f_0}, & 0 \le f_0 < 1/2 \\ e^{\mp j2\pi f_0}, & 1/2 \le f_0 < 1 \end{cases}$$

- and finally: $p_{1,2} = e^{\pm j2\pi f_0} \quad \Rightarrow \quad \left|p_{1,2}\right| = 1$

- <u>A sinusoidal parametric process with frequency f_0 can be seen as a degenerate AR(2) process with the poles on the unit circle and phases $\pm 2\pi f_0$</u> $\rightarrow$ the innovation filter is unstable and the innovation process has zero power.

Optimal 1-Step Linear Prediction: Cosinusoidal Process

- In general, a predictable process that has a PSD with L lines can be predicted with MSE=0 by means of a suitable linear predictor of order $P=L$, that is a FIR(L) filter.
- The samples of this process follow an homogeneous linear difference equation of order L. For this reason, the process can be seen as an AR(L) degenerate process with L poles on the unit circle. The phase of the k-th pole is $2\pi f_k$, where f_k is the frequency of the k-th line of the PSD.
- The process is not regular, since the innovation filter is unstable and the power of the innovation process is zero.

- In the following, we investigate the problem of deriving the 1-step optimal linear predictor (<u>without setting *a-priori* the filter order P</u>) for a predictable process and for a regular process, with *a-priori* known PSD.

Optimal 1-Step Linear Prediction: Predictable Processes

- **Optimal 1-step linear predictor for predictable processes**.
- The **complex Power Spectral Density (CPSD)** of a discrete-time **predictable** process *X*[*n*] is discrete; let's assume it contains *L* discrete lines:

$$R_X[m] = \sum_{i=1}^{L} P_i e^{j2\pi f_i m} \quad \overset{ZT}{\Leftrightarrow} \quad S_X(z) = \sum_{m=-\infty}^{+\infty} R_X[m] z^{-m} = \sum_{i=1}^{L} 2\pi P_i \delta\left(z - e^{j2\pi f_i}\right)$$

$$S_X\left(e^{j2\pi f}\right) = S_X(z)\Big|_{z=e^{j2\pi f}} = \sum_{i=1}^{L} 2\pi P_i \delta\left(e^{j2\pi f} - e^{j2\pi f_i}\right)$$

Optimal 1-Step Linear Prediction: Predictable Processes

- **Predictable processes**, whose PSD is discrete, do not satisfy the Paley-Wiener condition and can be predicted with MSE=0 by means of a predictor of a suitable order *P*.

- **Optimal 1-step linear predictor of order *P*:** we want to estimate $X[n]$ from $X[n-1]$, $X[n-2]$, … , $X[n-P]$:

$$\hat{X}[n] = \sum_{k=1}^{P} h[k]X[n-k] = h[1]X[n-1] + \cdots + h[P]X[n-P]$$

- The prediction error $\varepsilon[n]$ can be obtained by filtering $X[n]$ with a FIR filter called the **prediction error filter**, having impulse response $e[n]$:

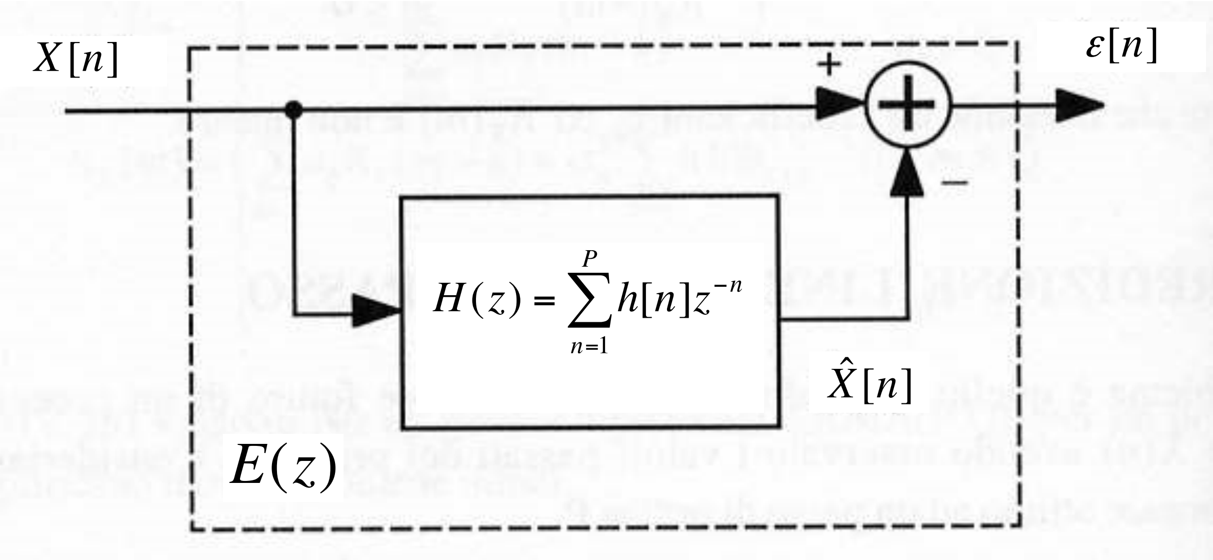
$$\varepsilon[n] = X[n] - \sum_{k=1}^{P} h[k]X[n-k] = e[n] \otimes X[n] \quad \Rightarrow \quad \varepsilon(z) = E(z)X(z)$$

Optimal 1-Step Linear Prediction: Predictable Processes

- The **prediction error filter** has system function *E*(*z*) given by:

$$\varepsilon(z) = E(z)X(z) = X(z) - H(z)X(z) = \left[1 - H(z)\right]X(z)$$

$$\Rightarrow \quad E(z) = 1 - H(z) = 1 - \sum_{k=1}^{P} h[k]z^{-k}$$



$$E\left(e^{j2\pi f}\right) = E(z)\Big|_{z=e^{j2\pi f}} = 1 - H\left(e^{j2\pi f}\right)$$

Optimal 1-Step Linear Prediction: Predictable Processes

- The power of the prediction error $\varepsilon[n]$, i.e. the prediction MSE:

$$\sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = \int_{-1/2}^{1/2} S_\varepsilon\left(e^{j2\pi f}\right)df = \int_{-1/2}^{1/2} S_X\left(e^{j2\pi f}\right)\left|E\left(e^{j2\pi f}\right)\right|^2 df$$

$$= \int_{-1/2}^{1/2} S_X\left(e^{j2\pi f}\right)\left|1 - H\left(e^{j2\pi f}\right)\right|^2 df$$

$$= \int_{-1/2}^{1/2} \left[\sum_{i=1}^{L} 2\pi P_i \delta\left(e^{j2\pi f} - e^{j2\pi f_i}\right)\right]\left|1 - H\left(e^{j2\pi f}\right)\right|^2 df$$

$$= \sum_{i=1}^{L} P_i \int_{-1/2}^{1/2} 2\pi\,\delta\left(e^{j2\pi f} - e^{j2\pi f_i}\right)\left|1 - H\left(e^{j2\pi f}\right)\right|^2 df$$

$$= \sum_{i=1}^{L} P_i \left|1 - H\left(e^{j2\pi f_i}\right)\right|^2 \geq 0$$

- We want to find $H(z)$ so that MSE=0.

Optimal 1-Step Linear Prediction: Predictable Processes

tutti non negativi

$$\sigma_\varepsilon^2 = \sum_{i=1}^{L} P_i \left|1 - H\left(e^{j2\pi f_i}\right)\right|^2 = 0$$

devo avere 0, in corrispondenza di quelle frequenze.

$$E\left(e^{j2\pi f_i}\right) = 1 - H\left(e^{j2\pi f_i}\right) = 0 \quad \text{for } i = 1, 2, \ldots, L$$

- In other words, the prediction error filter $E(z)$ must have L zeros on the unitary circle, for $z=z_i=\exp(j2\pi f_i)$, with $i=1,2,\ldots,L$, where L is the number of discrete lines in the PSD of $X[n]$ and $\{f_i\}$ are their frequencies.

MSE a 0

$$E(z) = \prod_{i=1}^{L}\left(1 - e^{j2\pi f_i} z^{-1}\right) \quad \Rightarrow \quad H(z) = 1 - E(z) = 1 - \prod_{i=1}^{L}\left(1 - e^{j2\pi f_i} z^{-1}\right)$$

Optimal 1-Step Linear Prediction: Predictable Processes

- The **optimal 1-step linear predictor** for a **predictable process** has order *P*=*L*, where *L* is the number of discrete lines in the PSD. It is a FIR(*L*) filter whose system function is:

$$H(z) = 1 - E(z) = 1 - \prod_{i=1}^{L}\left(1 - e^{j2\pi f_i} z^{-1}\right)$$

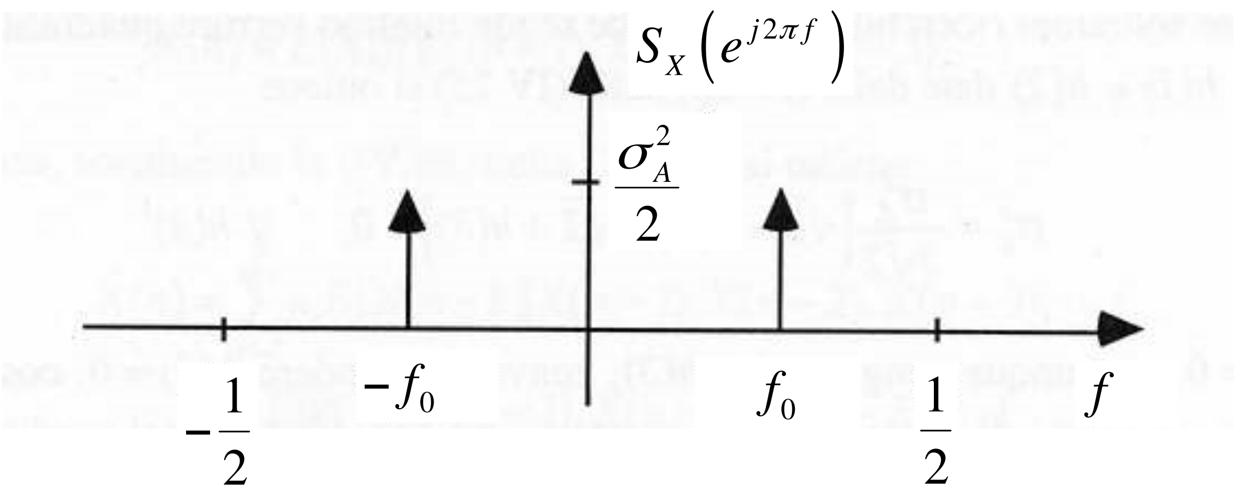
- ***Example***: Consider the problem of the 1-step optimal prediction of the parametric sinusoidal process previously considered. This process has two spectral lines in the PSD:

$$X[n] = A\cos\left[2\pi f_0 n + \theta\right], \quad A \in \mathcal{R}(\sigma_A^2), \quad \theta \in \mathcal{U}(-\pi, \pi)$$

Optimal 1-Step Linear Prediction: Predictable Processes

$$R_X[m] = \sigma_A^2 \cos\left[2\pi f_0 m\right]$$

$$S_X\left(e^{j2\pi f}\right) = 2\pi \cdot \frac{\sigma_A^2}{2}\left[\delta\left(e^{j2\pi f} - e^{j2\pi f_0}\right) + \delta\left(e^{j2\pi f} - e^{-j2\pi f_0}\right)\right]$$



- Line spectrum → predictable (not regular) process.

Optimal 1-Step Linear Prediction: Predictable Processes

$$H(z) = 1 - E(z) = 1 - \prod_{i=1}^{L}\left(1 - e^{j2\pi f_i} z^{-1}\right)$$

$$= 1 - \left(1 - e^{j2\pi f_0} z^{-1}\right)\left(1 - e^{-j2\pi f_0} z^{-1}\right)$$

$$= e^{j2\pi f_0} z^{-1} + e^{-j2\pi f_0} z^{-1} - z^{-2}$$

$$= 2\cos\left(2\pi f_0\right) z^{-1} - z^{-2}$$

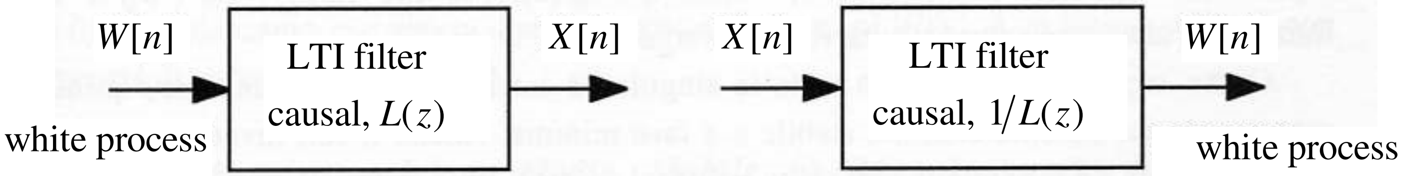
$$h[n] = Z^{-1}\left\{H(z)\right\} = 2\cos\left(2\pi f_0\right)\delta[n-1] - \delta[n-2]$$

$$\hat{X}[n] = h[n] \otimes X[n] = 2\cos\left(2\pi f_0\right) X[n-1] - X[n-2]$$

$$MSE\left\{\hat{X}[n]\right\} = 0 \quad \Rightarrow \quad \hat{X}[n] = X[n]$$

Optimal 1-Step Linear Prediction: Regular Processes

- **Regular processes** whose PSD satisfies the Paley-Wiener condition can be described through the **innovations representation**: per un IIR dovrei risolvere ∞ soluzioni



$$\sigma_W^2 = E\left\{W^2[n]\right\} \qquad X[n] = \sum_{k=0}^{+\infty} l[k]W[n-k]$$

- **Innovation filter**
- **Whitening filter**

$$l[n] \equiv \text{impulse response of the innovation filter,} \quad l[0] = \lim_{z\to\infty} L(z) = 1$$

- The regular process $X[n]$ and its innovation process $W[n]$ are **linearly equivalent**, i.e. we can transform one into the other by means of a LTI filter that is causal and stable with inverse causal and stable.

Optimal 1-Step Linear Prediction: Regular Processes

$$X[n] = l[n] \otimes W[n] = \sum_{k=0}^{+\infty} l[k] W[n-k]$$

$$\left. \begin{aligned} R_W[m] &= E\{W[n]W[n+m]\} = \sigma_W^2 \delta[m] \\ S_W\left(e^{j2\pi f}\right) &= \sum_{m=-\infty}^{+\infty} R_W[m] e^{-j2\pi fm} = \sigma_W^2 \end{aligned} \right\} \quad W[n] \text{ innovation process}$$

$$S_X(z) = \sigma_W^2 L(z) L(1/z) \quad \Rightarrow \quad S_X\left(e^{j2\pi f}\right) = \sigma_W^2 \left| L\left(e^{j2\pi f}\right) \right|^2$$

$$\Gamma(z) = \frac{1}{L(z)} \equiv \text{system function of the whitening filter}$$

$$\gamma[n] \equiv \text{impulse response of the whitening filter}$$

Optimal 1-Step Linear Prediction: Regular Processes

- **Optimal 1-step linear predictor** (of order P eventually $+\infty$): our aim is to estimate $X[n]$ from the observations $X[n\text{-}1]$, $X[n\text{-}2]$, $x[n\text{-}3]$, …

$$\hat{X}[n] = \sum_{k=1}^{+\infty} h[k]X[n-k]$$

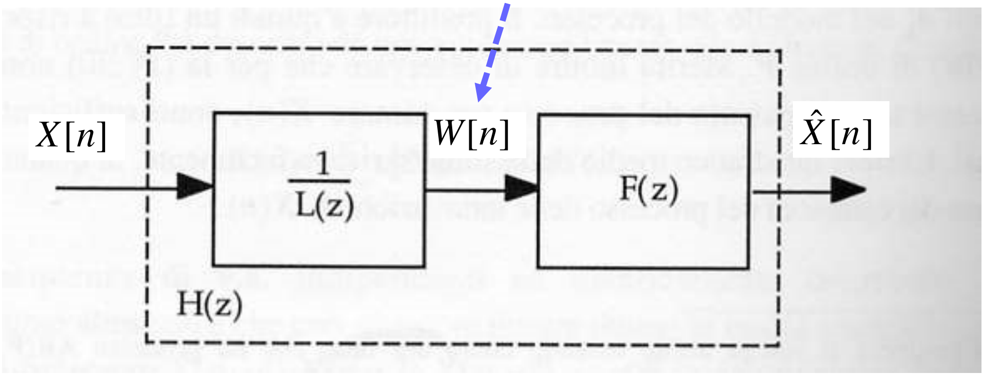
- Since the regular process $X[n]$ and its innovation process $W[n]$ are **linearly equivalent**:

$$X[n] = l[n] \otimes W[n] = \sum_{k=0}^{+\infty} l[k]W[n-k] = l[0]W[n] + \sum_{k=1}^{+\infty} l[k]W[n-k]$$

- we can express the optimal linear predictor as the cascade of a whitening filter and of the optimal linear predictor that processes the whitened data, i.e. the samples of $W[n]$.

Optimal 1-Step Linear Prediction: Regular Processes

$$W[n] = \gamma[n] \otimes X[n] = \sum_{k=0}^{+\infty} \gamma[k] X[n-k]$$



$$\hat{X}[n] = \sum_{k=1}^{+\infty} h[k] X[n-k] = \sum_{k=1}^{+\infty} f[k] W[n-k]$$

- The optimal 1-step linear prediction that processes the whitened data can be derived by imposing the orthogonality between the error ε[*n*] and the data *W*[*n*-1], *W*[*n*-2], *W*[*n*-3], … :

$$E\left\{\varepsilon[n] W[n-m]\right\} = E\left\{\left(X[n] - \hat{X}[n]\right) W[n-m]\right\} = 0, \quad m \geq 1$$

$$E\left\{\left(X[n]-\hat{X}[n]\right)W[n-m]\right\}$$

$$=E\left\{\left(\sum_{k=0}^{+\infty} l[k]W[n-k]-\sum_{k=1}^{+\infty} f[k]W[n-k]\right)W[n-m]\right\}$$

$$=E\left\{\left(l[0]W[n]+\sum_{k=1}^{+\infty}\left(l[k]-f[k]\right)W[n-k]\right)W[n-m]\right\}$$

$$=l[0]R_W[m]+\sum_{k=1}^{+\infty}\left(l[k]-f[k]\right)R_W[m-k]$$

$$=\sigma_W^2\delta[m]+\sigma_W^2\sum_{k=1}^{+\infty}\left(l[k]-f[k]\right)\delta[m-k]=0,\quad m\geq 1$$

→ infinite incognite in infinite eq

↓

Tuttavia sono disaccoppiate

$$\sum_{k=1}^{+\infty}\left(l[k]-f[k]\right)\delta[m-k]=0,\quad m\geq 1$$

Optimal 1-Step Linear Prediction: Regular Processes

- The solution of the linear system we derived is trivially given by:

$$f[m] = \begin{cases} l[m], & m \geq 1 \\ 0, & otherwise \end{cases}$$

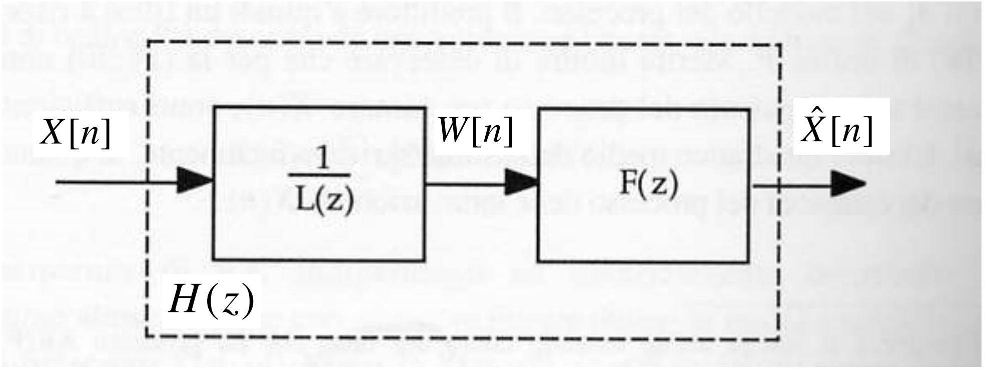
- The system function $F(z)$ of the optimal 1-step linear predictor that processes the whitened data is given by the one-sided Z-transform of $f[n]$:

$$F(z) = \sum_{n=0}^{+\infty} f[n] z^{-n} = \sum_{n=1}^{+\infty} l[n] z^{-n} = \sum_{n=0}^{+\infty} l[n] z^{-n} - l[0] = L(z) - 1$$

- where $L(z)$ is the system function of the innovation filter:

$$L(z) = \sum_{n=0}^{\infty} l[n] z^{-n} = 1 + \sum_{n=1}^{\infty} l[n] z^{-n}$$

Optimal 1-Step Linear Prediction: Regular Processes



- The optimal 1-step linear predictor that processes the samples of the observed process $X[n]$ is given by the cascade of the whitening filter and of the optimal 1-step linear predictor that processes the whitened data. Consequently, its system function can be expressed as:

$$H(z) = \frac{F(z)}{L(z)} = \frac{L(z)-1}{L(z)} = 1 - \frac{1}{L(z)} = 1 - \Gamma(z)$$

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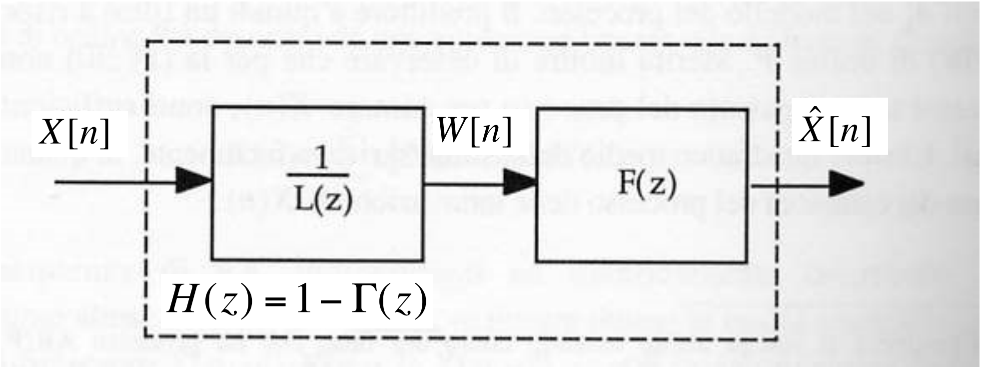

- The impulse response of the optimal 1-step predictor is then:

$$H(z) = 1 - \Gamma(z) \quad \Rightarrow \quad h[n] = \begin{cases} 0, & n = 0 \\ -\gamma[n], & n \geq 1 \end{cases}$$

$$h[0] = \lim_{z \to \infty} H(z) = 1 - \lim_{z \to \infty} \Gamma(z) = 1 - \lim_{z \to \infty} \frac{1}{L(z)} = 0$$

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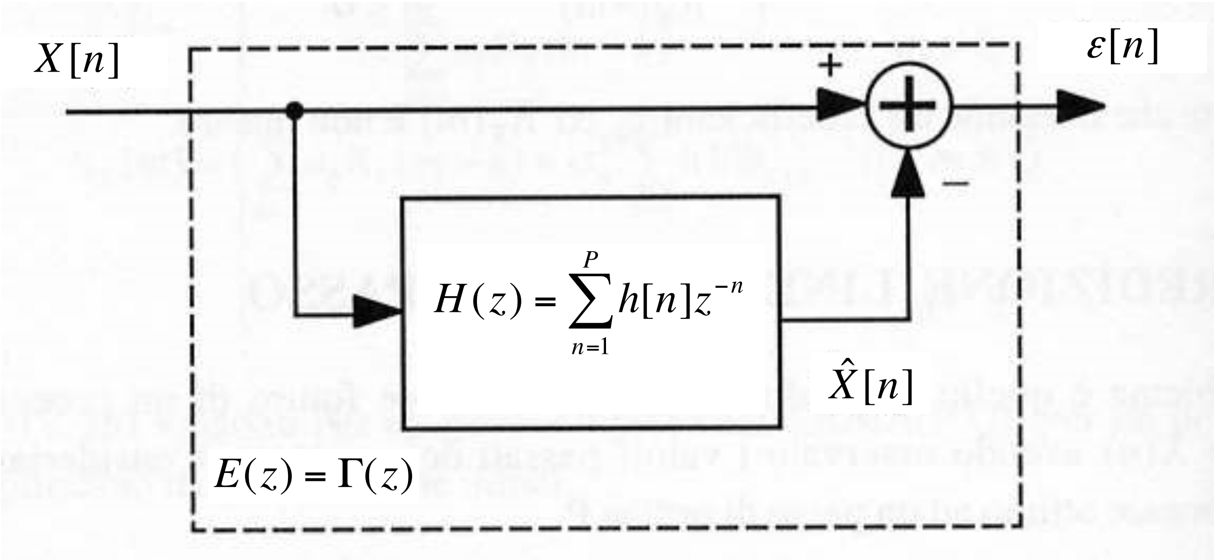
- The **MSE** can be obtained by representing the prediction error as the output of the prediction error filter, that has system function $E(z)=1-H(z)$:

$$E(z) = 1 - H(z) = 1 - \left(1 - \Gamma(z)\right) \quad \Rightarrow \quad E(z) = \Gamma(z)$$



- The **prediction error filter** is equal to the **whitening filter**, then $\varepsilon[n]=W[n]$, and their powers are equal.

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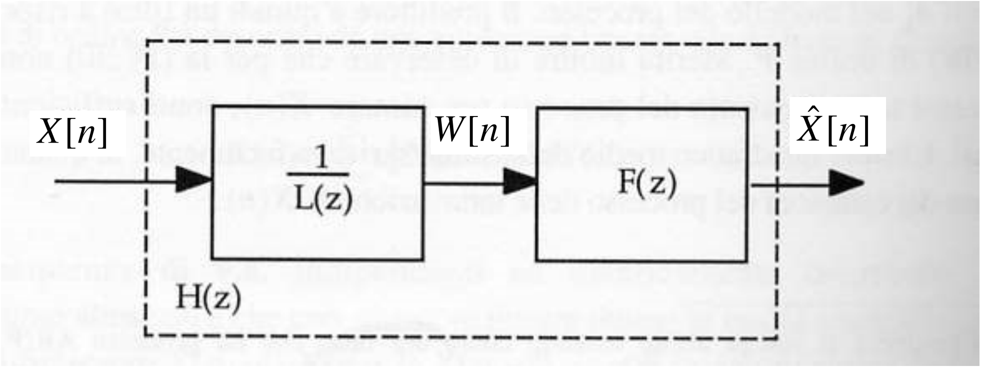


$$
\begin{cases}
H(z) = 1 - \Gamma(z) \\[2em]
h[n] = \begin{cases} 0, & n \le 0 \\ -\gamma[n], & n \ge 1 \end{cases} \\[2em]
MSE\left\{\hat{X}[n]\right\} = \sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{W^2[n]\right\} = \sigma_W^2
\end{cases}
$$

Optimal 1-Step Linear Prediction: Regular Processes

- **Example**: $X[n]$ is a **MA(Q) process**.

$$X[n] = \sum_{k=0}^{Q} b_k W[n-k], \quad b_0 = 1$$

$$L(z) = 1 + \sum_{k=1}^{Q} b_k z^{-k} \quad \Rightarrow \quad \Gamma(z) = \frac{1}{1 + \sum_{k=1}^{Q} b_k z^{-k}}$$

$$\begin{cases} H(z) = 1 - \Gamma(z) = 1 - \dfrac{1}{1 + \sum_{k=1}^{Q} b_k z^{-k}} = \dfrac{\sum_{k=1}^{Q} b_k z^{-k}}{1 + \sum_{k=1}^{Q} b_k z^{-k}} \\ MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 \end{cases}$$

Optimal 1-Step Linear Prediction: Regular Processes

- **Example**: $X[n]$ is a **MA(1) process**.

$$X[n] = W[n] + b_1 W[n-1], \quad |b_1| < 1$$

- Let us first derive the ACF and the PSD. The ACF can be derived as follows:

$$R_X[m] = E\{X[n]X[n+m]\}$$

$$= E\{(W[n] + b_1 W[n-1])(W[n+m] + b_1 W[n+m-1])\}$$

$$= R_W[m] + b_1 R_W[m-1] + b_1 R_W[m+1] + b_1^2 R_W[m]$$

$$= (1 + b_1^2)\sigma_W^2 \delta[m] + b_1 \sigma_W^2 \delta[m-1] + b_1 \sigma_W^2 \delta[m+1]$$

Optimal 1-Step Linear Prediction: Regular Processes

$$R_X[m] = \left(1 + b_1^2\right)\sigma_W^2 \delta[m] + b_1 \sigma_W^2 \delta[m-1] + b_1 \sigma_W^2 \delta[m+1]$$

$$R_X[m] = \begin{cases} \left(1 + b_1^2\right)\sigma_W^2, & m = 0 \\ b_1 \sigma_W^2, & m = \pm 1 \\ 0, & otherwise \end{cases}$$

- The power of the process and the one-lag correlation coefficient are given by:

$b_1 > 0 \to$ passabasso

$b_1 < 0 \to$ passaalto

$$R_X[0] = \sigma_X^2 = \left(1 + b_1^2\right)\sigma_W^2, \qquad \rho_X[1] = \frac{R_X[1]}{R_X[0]} = \frac{b_1}{1 + b_1^2}$$

banda dim

$b_1 \to 1$

Optimal 1-Step Linear Prediction: Regular Processes

- The PSD can be derived as the Fourier Transform (FT) of the ACF:

$$S_X\left(e^{j2\pi f}\right) = FT\left\{R_X[m]\right\} = \sum_{m=-\infty}^{+\infty} R_X[m]e^{-j2\pi fm}$$

$$= \sum_{m=-\infty}^{+\infty} \left[\left(1+b_1^2\right)\sigma_W^2\delta[m] + b_1\sigma_W^2\delta[m-1] + b_1\sigma_W^2\delta[m+1]\right]e^{-j2\pi fm}$$

$$= \left(1+b_1^2\right)\sigma_W^2 + b_1\sigma_W^2 e^{-j2\pi f} + b_1\sigma_W^2 e^{j2\pi f}$$

$$= \sigma_W^2\left(1+b_1^2+2b_1\cos\left(2\pi f\right)\right)$$ → f. reale e pari

$$b_1 < 0 \quad \Rightarrow \quad \text{high-pass process}, \qquad b_1 > 0 \quad \Rightarrow \quad \text{low-pass process}$$

Optimal 1-Step Linear Prediction: Regular Processes

- The PSD can also be obtained by deriving first the innovation filter:

$$X[n] = W[n] + b_1 W[n-1] \quad \Rightarrow \quad X(z) = W(z) + b_1 z^{-1} W(z)$$

- Hence, the innovation filter for an MA(1) process is given by:

$$L(z) = \frac{X(z)}{W(z)} = 1 + b_1 z^{-1} = \frac{z + b_1}{z}$$

- The innovation filter has a zero in z=-b_1 and a pole in z=0.
- Then, the PSD:

$$S_X\left(e^{j2\pi f}\right) = \sigma_W^2 \left|L\left(e^{j2\pi f}\right)\right|^2 = \sigma_W^2 \left|1 + b_1 e^{-j2\pi f}\right|^2$$

$$= \sigma_W^2 \left(1 + b_1^2 + 2b_1 \cos\left(2\pi f\right)\right)$$

Optimal 1-Step Linear Prediction: Regular Processes

- The whitening filter is an IIR(1) filter given by:

$$\Gamma(z) = \frac{1}{L(z)} = \frac{1}{1 + b_1 z^{-1}}$$

- The whitening filter has a pole in $z=-b_1$ (that should be inside the unit circle for the stability) and a zero in $z=0$.

- The 1-step optimal linear predictor is an IIR(1) filter with system function:

$$H(z) = 1 - \Gamma(z) = 1 - \frac{1}{1 + b_1 z^{-1}} = \frac{b_1 z^{-1}}{1 + b_1 z^{-1}}$$

Optimal 1-Step Linear Prediction: Regular Processes

- The frequency response of the 1-step optimal predictor is given by:

$$H\left(e^{j2\pi f}\right) = H(z)\Big|_{z=e^{j2\pi f}} = \frac{b_1 e^{-j2\pi f}}{1 + b_1 e^{-j2\pi f}} = \frac{b_1}{e^{j2\pi f} + b_1}$$

$$\left|H\left(e^{j2\pi f}\right)\right| = \frac{|b_1|}{\sqrt{\left(\cos\left(2\pi f\right) + b_1\right)^2 + \sin^2\left(2\pi f\right)}}$$

$$= \frac{|b_1|}{\sqrt{1 + b_1^2 + 2b_1 \cos\left(2\pi f\right)}}$$

$b_1 < 0 \quad \Rightarrow \quad$ high-pass process, $\quad b_1 > 0 \quad \Rightarrow \quad$ low-pass process

- By evaluating the inverse Z-transform of the linear predictor system function, we obtain its impulse response:

$\sum_{0}^{\infty} x^n = \frac{1}{1-x}, \quad |x|<1$

→ serie geometrica

$$H(z) = \frac{b_1 z^{-1}}{1+b_1 z^{-1}} = b_1 z^{-1} \sum_{k=0}^{+\infty} \left(-b_1 z^{-1}\right)^k = -\sum_{k=0}^{+\infty} \left(-b_1\right)^{k+1} z^{-(k+1)}$$

$$= -\sum_{n=1}^{+\infty} \left(-b_1\right)^n z^{-n} \quad \Rightarrow \quad h[n] = \begin{cases} -\left(-b_1\right)^n, & n \geq 1 \\ 0, & otherwise \end{cases}$$

$$\hat{X}[n] = \sum_{k=1}^{+\infty} h[k] X[n-k] = -\sum_{k=1}^{+\infty} \left(-b_1\right)^k X[n-k]$$

Optimal 1-Step Linear Prediction: Regular Processes

- The optimal 1-step linear predictor can be expressed in recursive form, which is more efficient from a computational point of view, as follows:

$$H(z) = \frac{b_1 z^{-1}}{1 + b_1 z^{-1}} \quad \Rightarrow \quad \hat{X}(z) = H(z)X(z) = \frac{b_1 z^{-1} X(z)}{1 + b_1 z^{-1}}$$

$$\Rightarrow \quad \left(1 + b_1 z^{-1}\right)\hat{X}(z) = b_1 z^{-1} X(z) \quad \Rightarrow \quad \hat{X}(z) = -b_1 z^{-1} \hat{X}(z) + b_1 z^{-1} X(z)$$

- By applying the inverse Z-transform, we get:

$$\begin{cases} \hat{X}[n] = -b_1 \hat{X}[n-1] + b_1 X[n-1] = b_1 \left(X[n-1] - \hat{X}[n-1] \right) \\ MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 = \dfrac{\sigma_X^2}{1 + b_1^2} \end{cases}$$

Optimal 1-Step Linear Prediction: Regular Processes

- We could have derived the same result following a different reasoning, under the assumption that the innovation process in addition to be white, is also an IID process:

$$X[n] = W[n] + b_1 W[n-1]$$

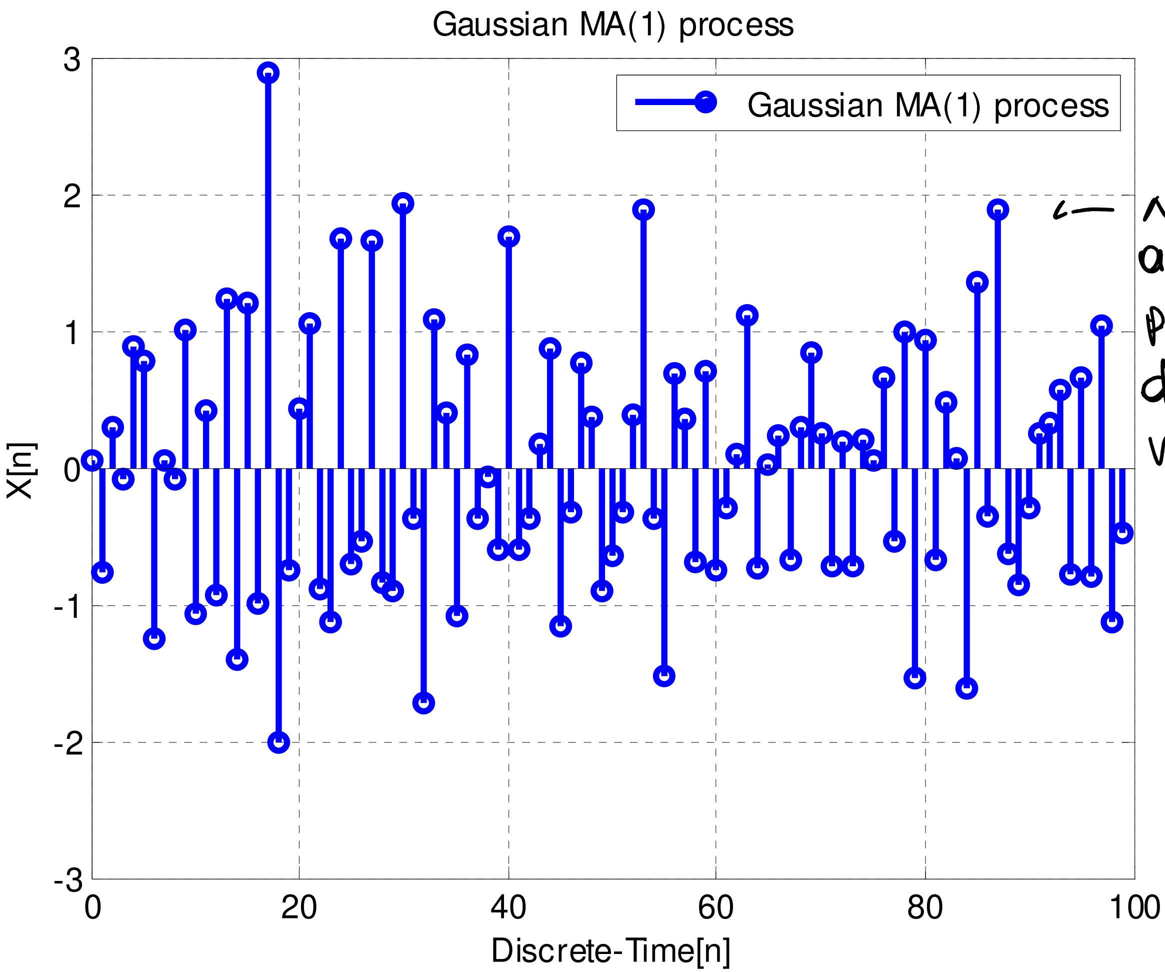
$$\begin{aligned}
\hat{X}[n] &= E\left\{X[n] \middle| X[n-1], X[n-2], \cdots\right\} = E\left\{X[n] \middle| W[n-1], W[n-2], \cdots\right\} \\
&= E\left\{W[n] + b_1 W[n-1] \middle| W[n-1], W[n-2], \cdots\right\} \\
&= E\left\{W[n] \middle| W[n-1], W[n-2], \cdots\right\} + E\left\{b_1 W[n-1] \middle| W[n-1], W[n-2], \cdots\right\} \\
&= E\left\{W[n]\right\} + b_1 W[n-1] \\
&= b_1 W[n-1] \\
&= b_1\left(X[n-1] - \hat{X}[n-1]\right)
\end{aligned}$$

Optimal 1-Step Linear Prediction: Regular Processes

b

< 0 HP a banda abbastanza larga

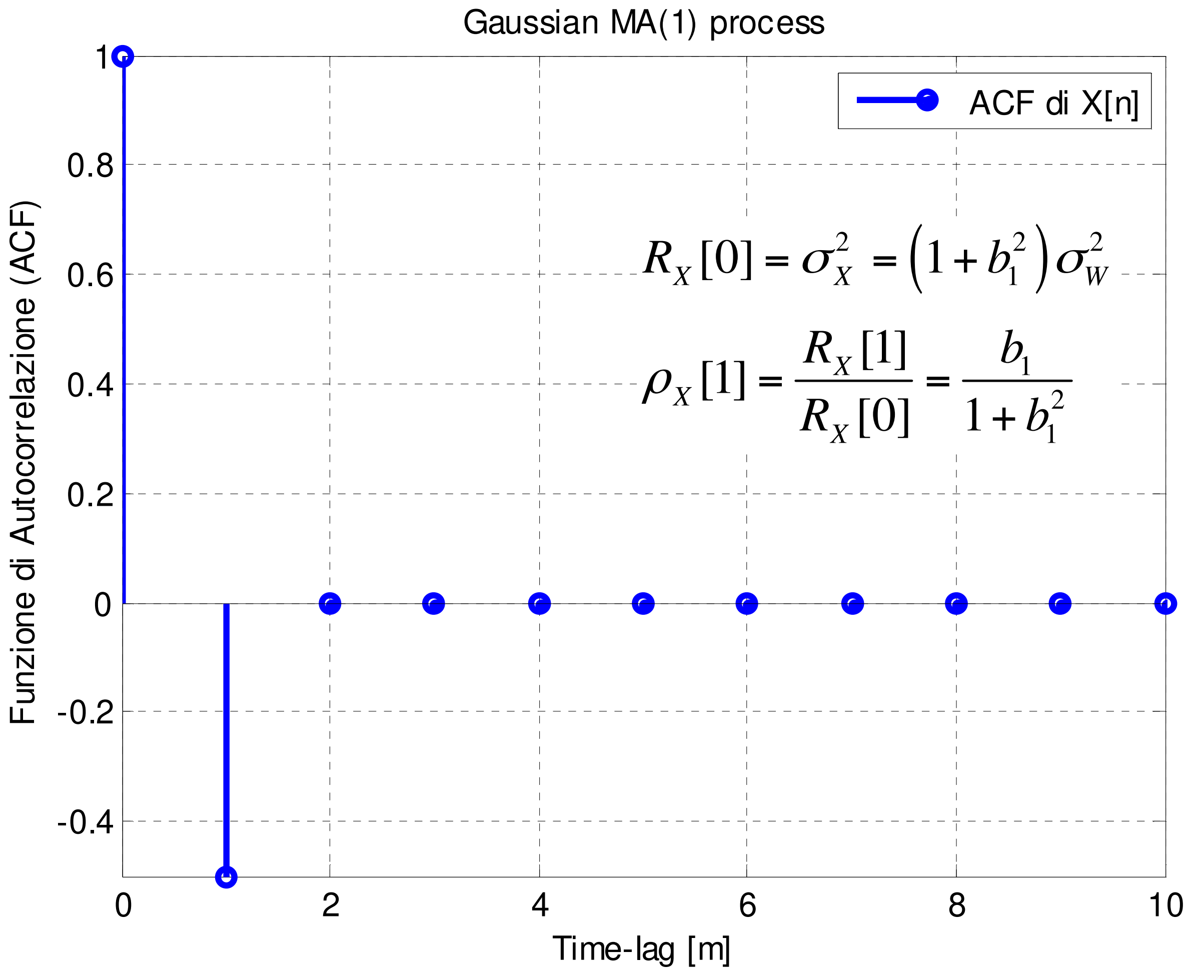
$$b_1 = -0.98, \quad z_1 = 0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ high-pass process}$$



No componenti a b. frequenza per alternanza di segno nei valori.

Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = -0.98, \quad z_1 = 0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ high-pass process}$$

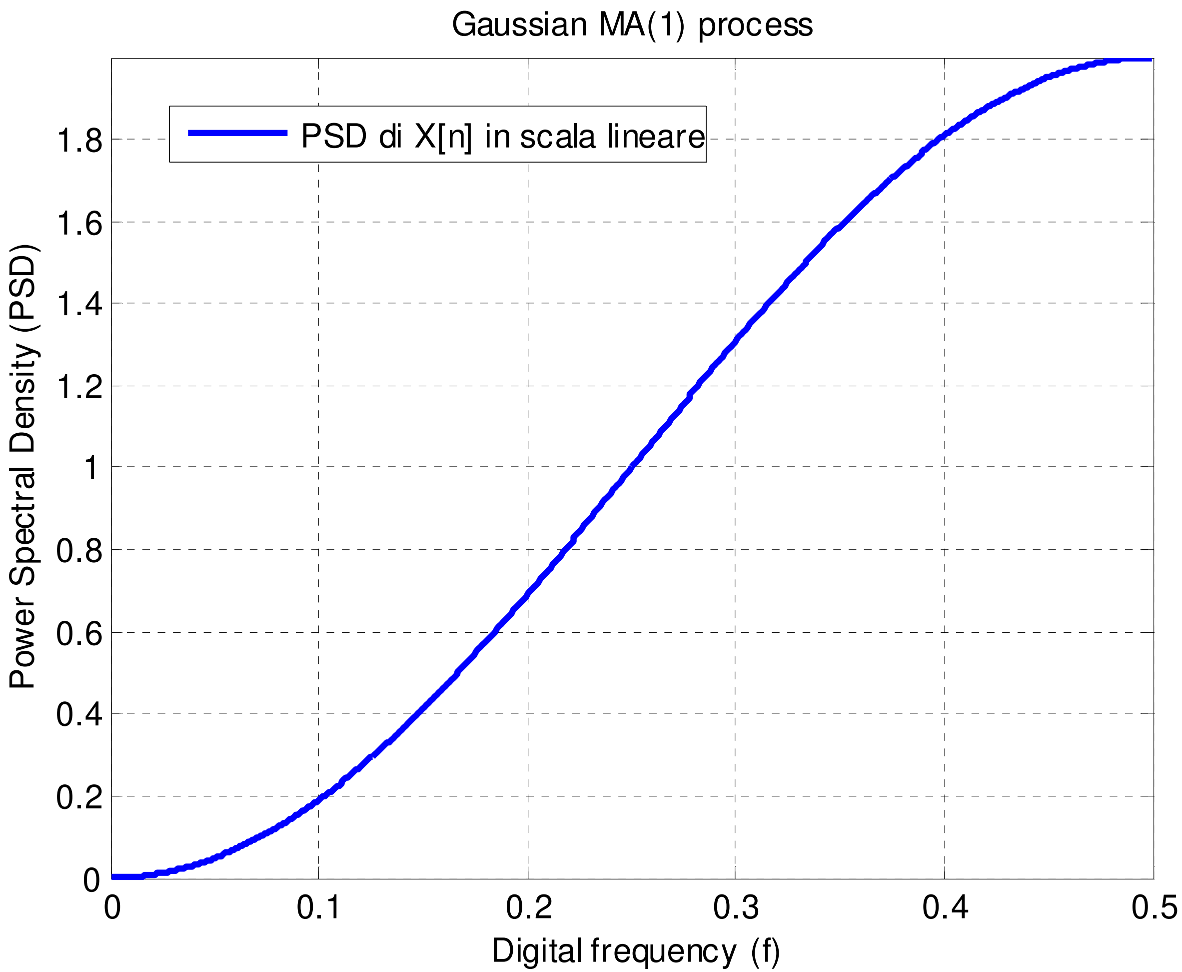


$$R_X[0] = \sigma_X^2 = \left(1 + b_1^2\right)\sigma_W^2$$

$$\rho_X[1] = \frac{R_X[1]}{R_X[0]} = \frac{b_1}{1 + b_1^2}$$

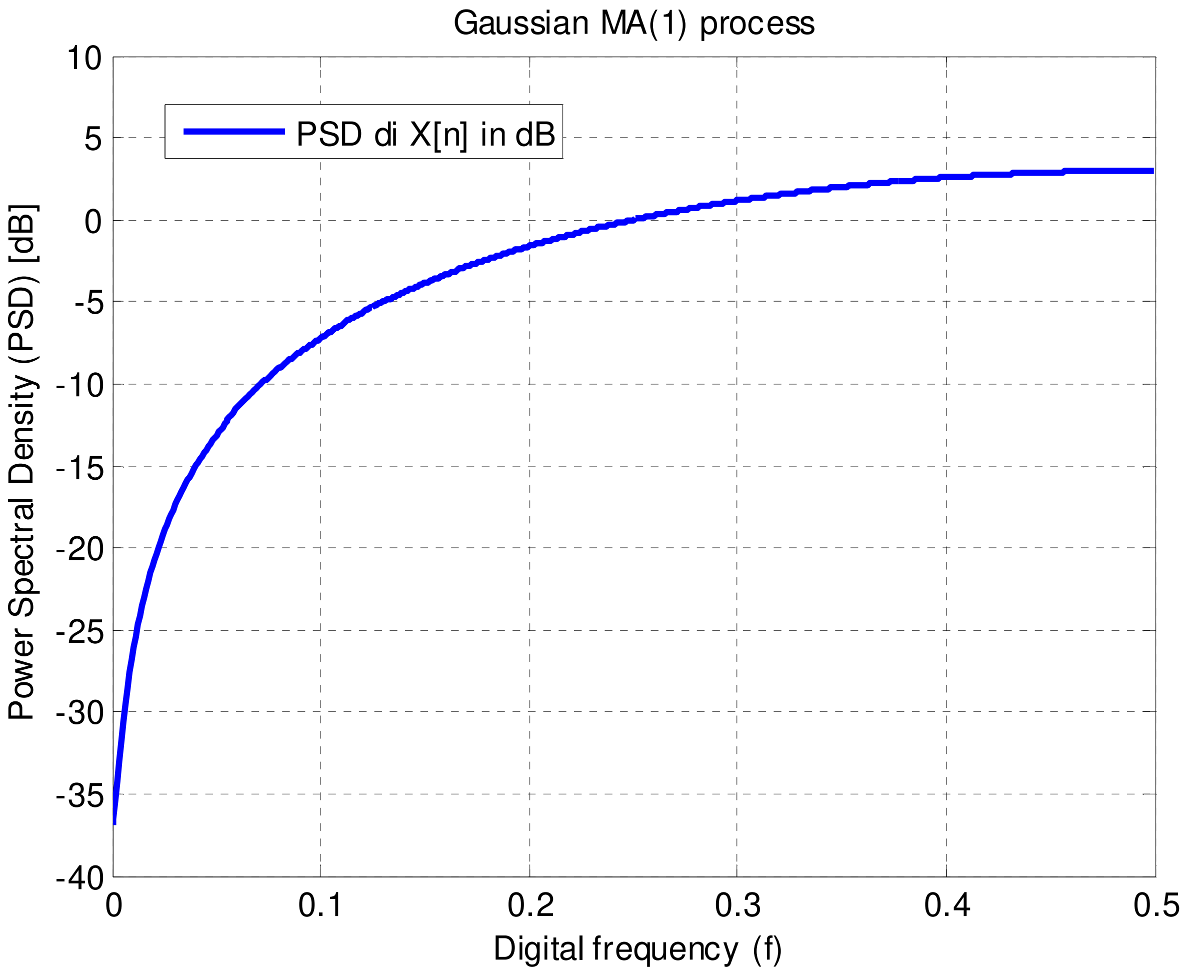
Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = -0.98, \quad z_1 = 0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ high-pass process}$$



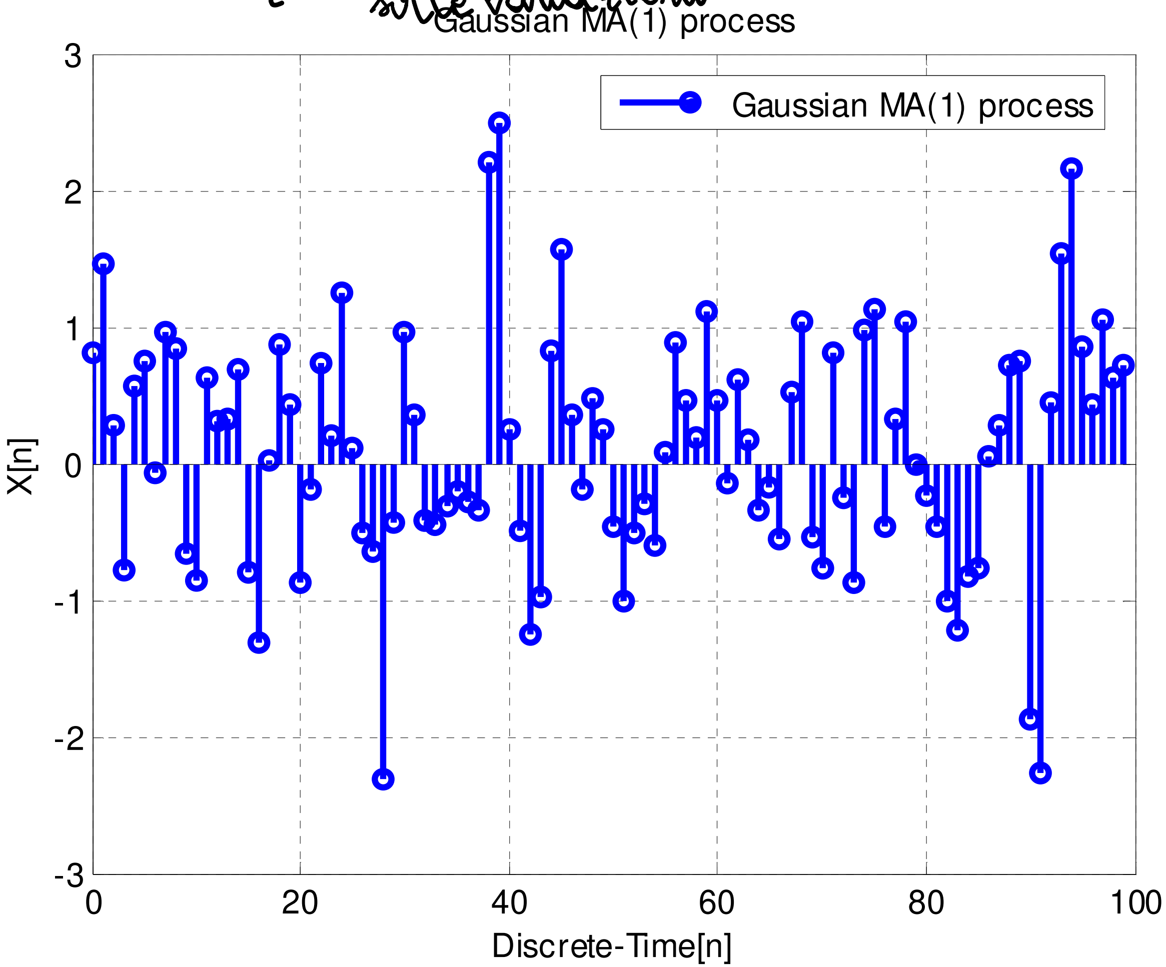
Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = -0.98, \quad z_1 = 0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ high-pass process}$$



$$b_1 = 0.98, \quad z_1 = -0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ low-pass process}$$

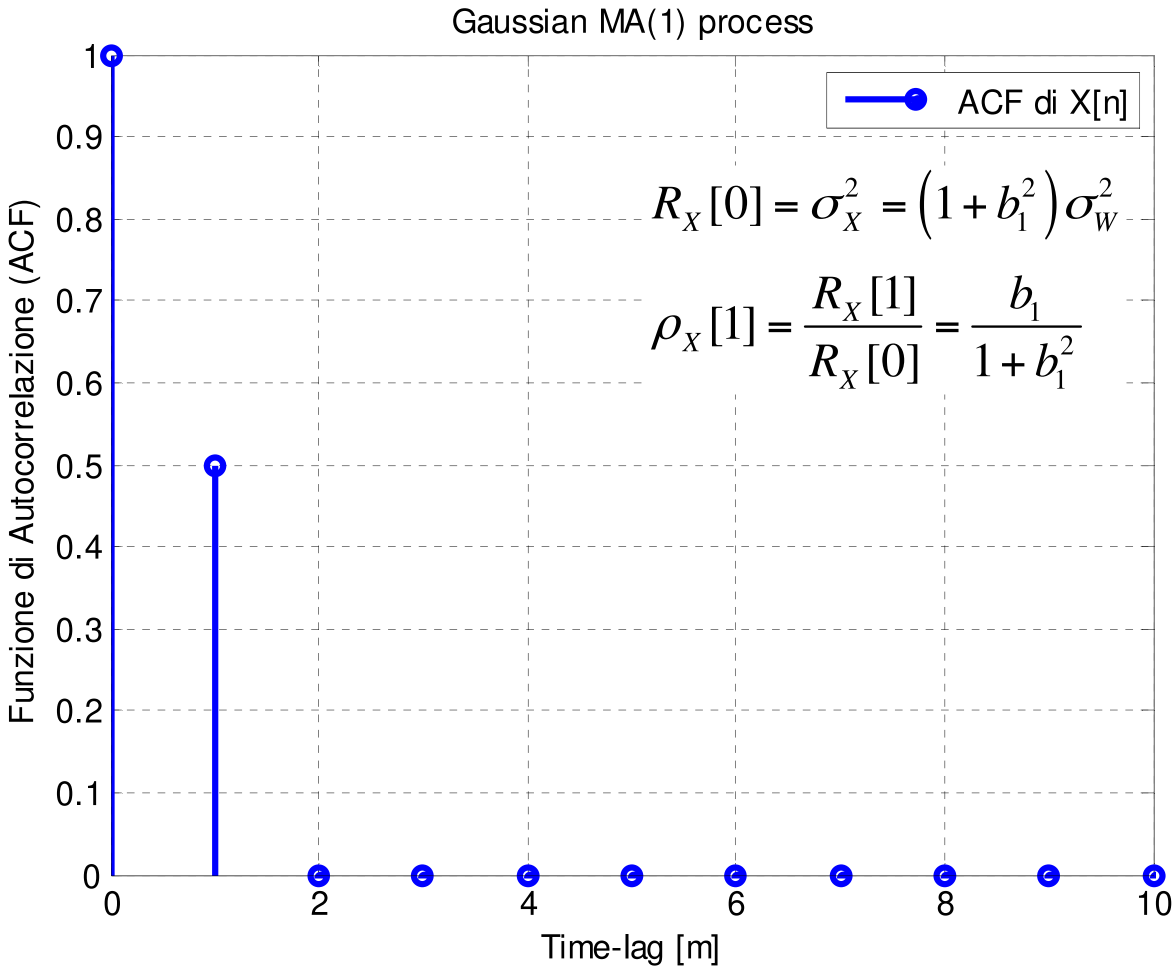
buco $f = 1/2$, passa-basso → banda larga · ha detto roba
succe variazioni



Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = 0.98, \quad z_1 = -0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ low-pass process}$$

$$\rho = \frac{b_1}{1+b_1^2}$$

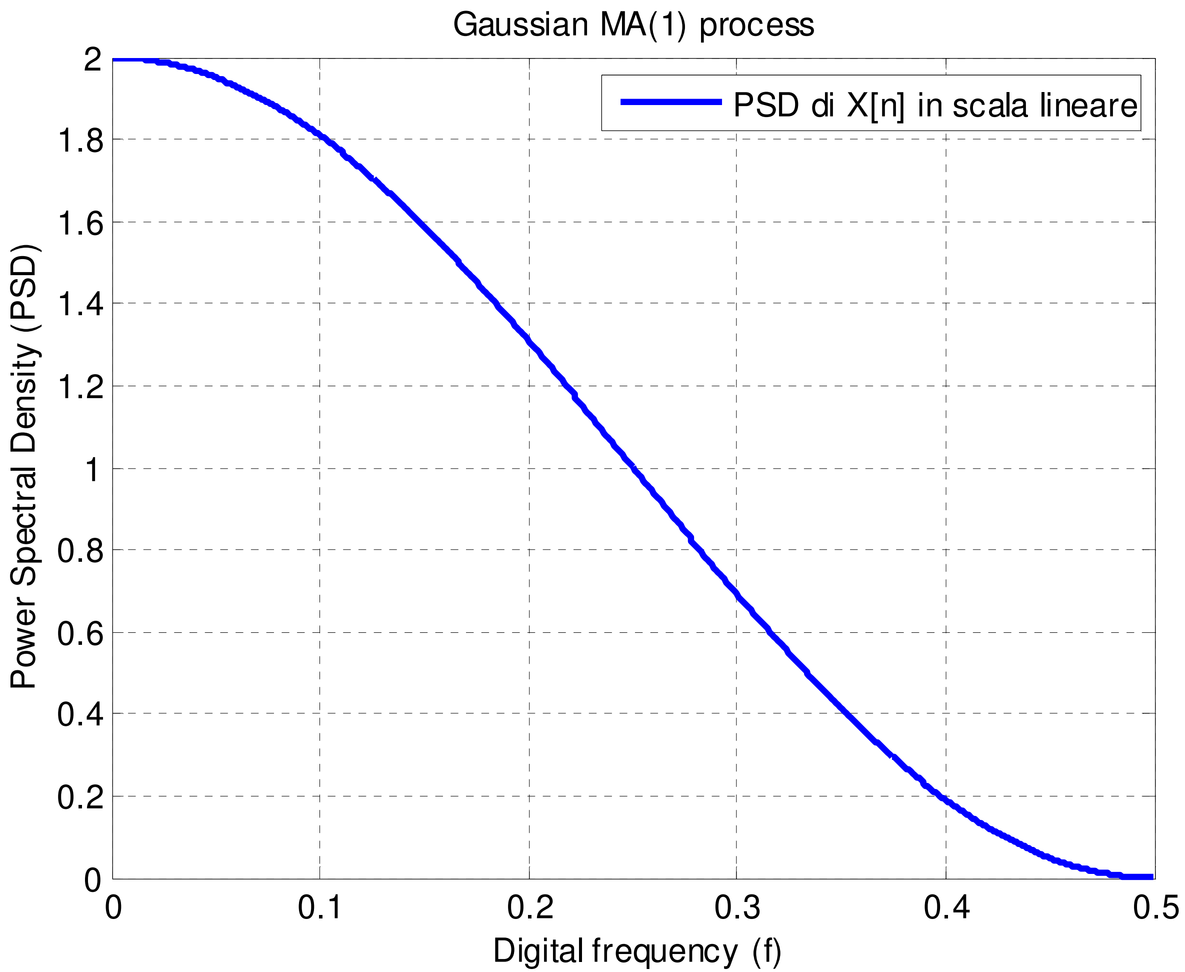


$$R_X[0] = \sigma_X^2 = \left(1 + b_1^2\right)\sigma_W^2$$

$$\rho_X[1] = \frac{R_X[1]}{R_X[0]} = \frac{b_1}{1 + b_1^2}$$

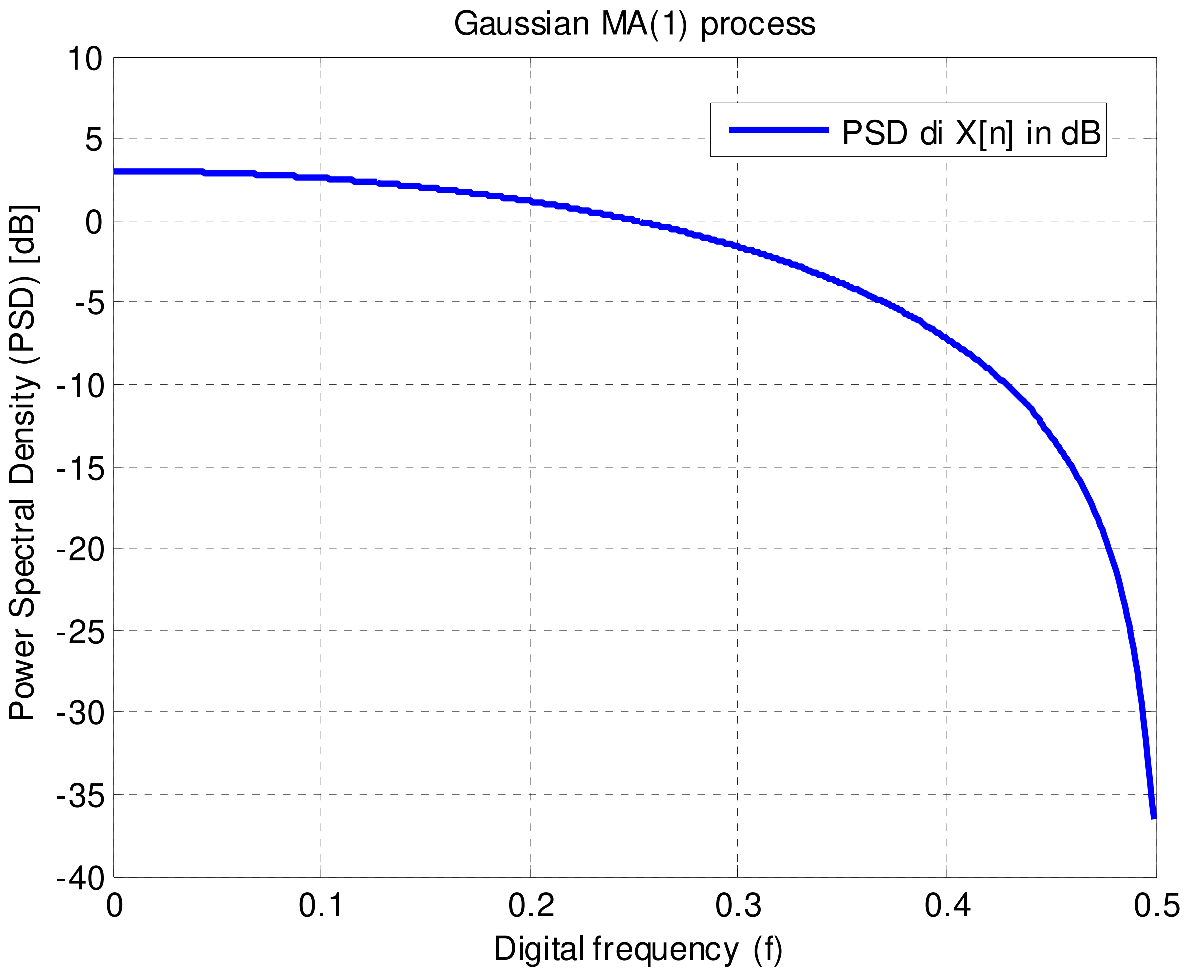
Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = 0.98, \quad z_1 = -0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ low-pass process}$$



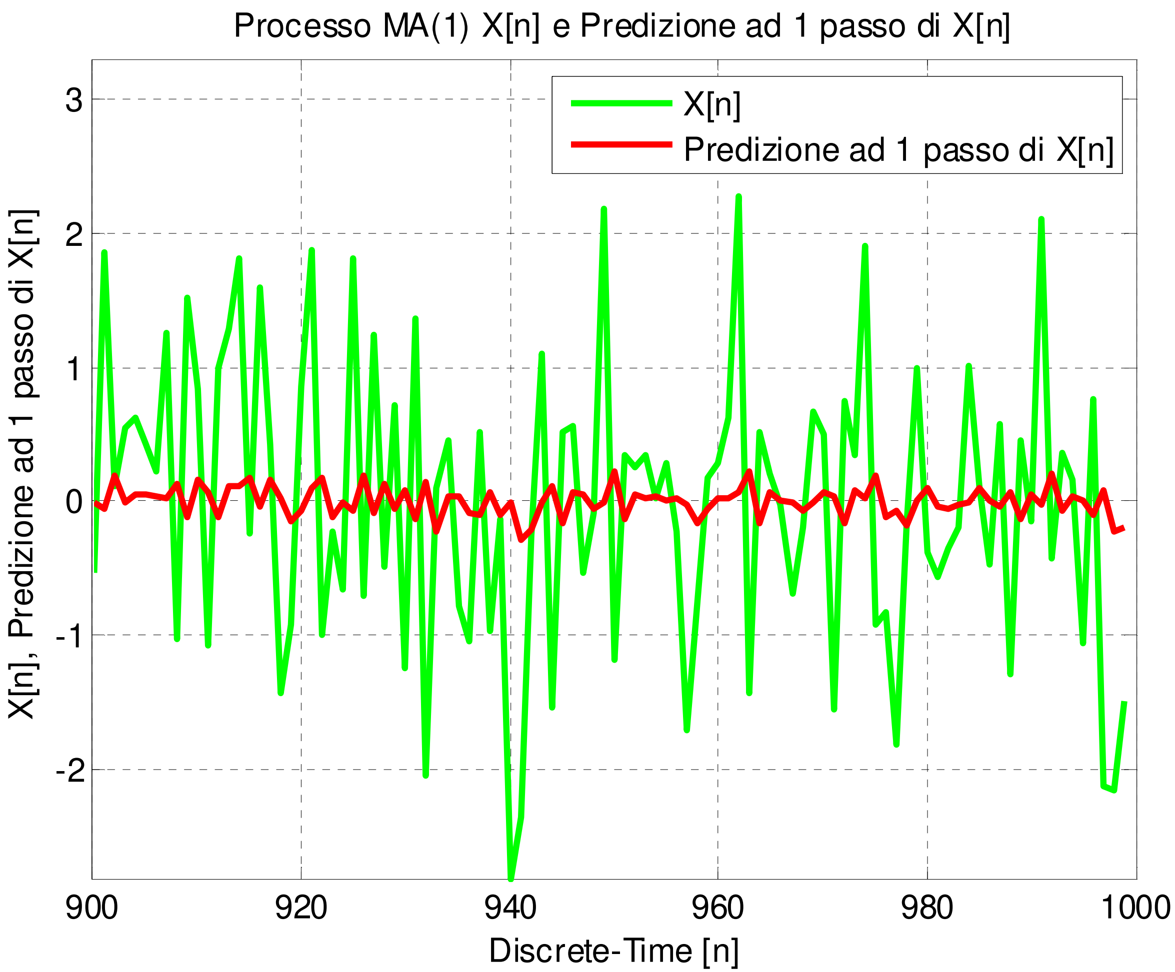
Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = 0.98, \quad z_1 = -0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad X[n] \text{ low-pass process}$$



Optimal 1-Step Linear Prediction: Regular Processes

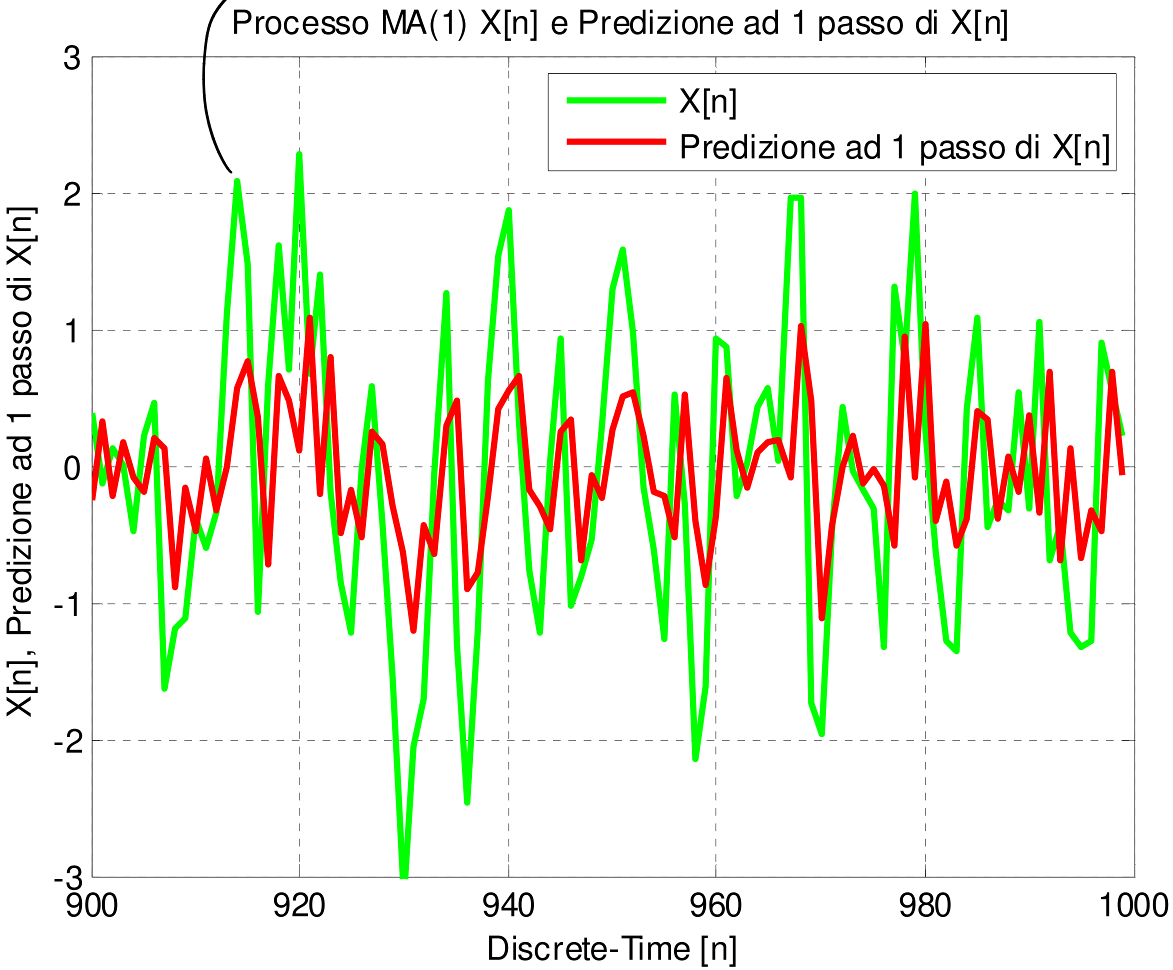
$$b_1 = 0.1, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 = 0.99$$



Optimal 1-Step Linear Prediction: Regular Processes

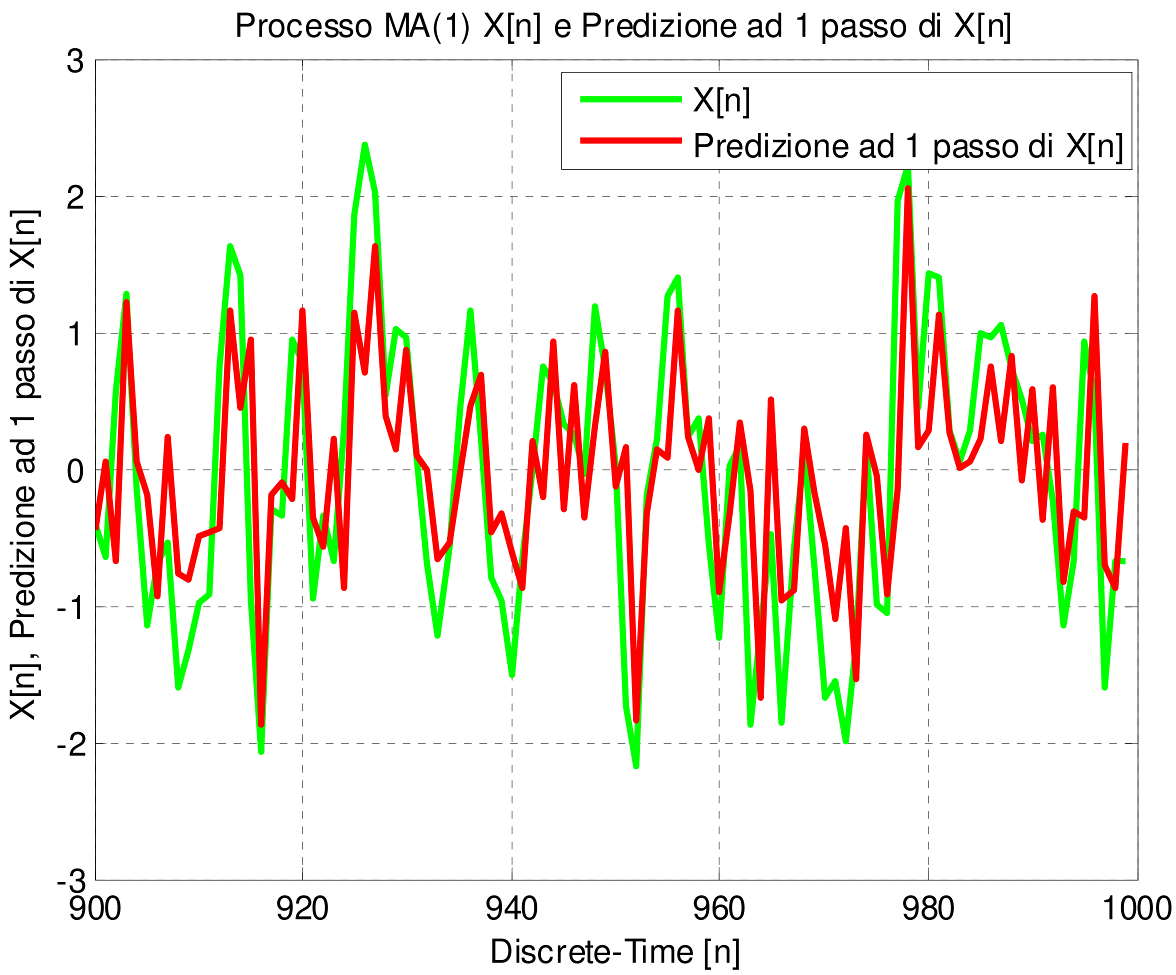
$$b_1 = 0.5, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 = 0.80$$

→ minor var. → si stringe banda → migliora predizione



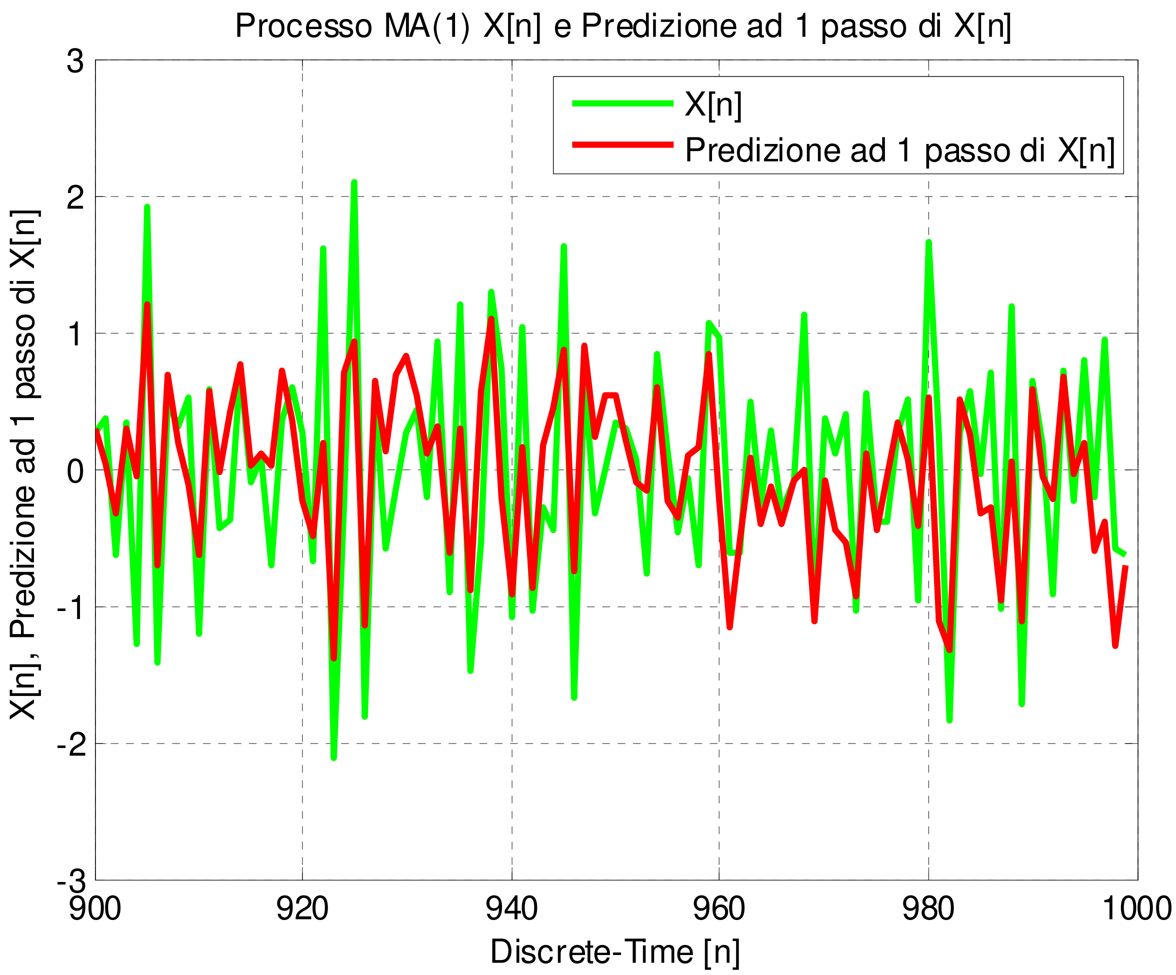
Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = 0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 = 0.51$$



Optimal 1-Step Linear Prediction: Regular Processes

$$b_1 = -0.98, \quad \sigma_X^2 = 1 \quad \Rightarrow \quad MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 = 0.51$$



Optimal 1-Step Linear Prediction: Regular Processes

- **Example**: *X*[*n*] is an **AR(*P*) process**.

$$X[n] = \sum_{k=1}^{P} a_k X[n-k] + W[n]$$

$$X(z) = \sum_{k=1}^{P} a_k X(z) z^{-k} + W(z)$$

$$X(z)\left(1 - \sum_{k=1}^{P} a_k z^{-k}\right) = W(z)$$

$$L(z) = \frac{X(z)}{W(z)} = \frac{1}{1 - \sum_{k=1}^{P} a_k z^{-k}} \quad \Rightarrow \quad \Gamma(z) = \frac{1}{L(z)} = 1 - \sum_{k=1}^{P} a_k z^{-k}$$

Optimal 1-Step Linear Prediction: Regular Processes

$$\begin{cases} H(z) = 1 - \Gamma(z) = \sum_{k=1}^{P} a_k z^{-k} \\ h[n] = \begin{cases} a_n, & 1 \le n \le P \\ 0, & otherwise \end{cases} \\ MSE\left\{\hat{X}[n]\right\} = \sigma_W^2 \end{cases}$$

$$\hat{X}[n] = \sum_{k=1}^{P} a_k X[n-k]$$

Optimal 1-Step Linear Prediction: AR(3) Process

- ***Example***: Let $X[n]$ be an **AR(3) process**, such that:

$$X[n] = \sum_{k=1}^{3} a_k X[n-k] + W[n] \quad \text{with} \quad a_1 = 4/5, a_2 = 1/4, a_3 = -1/5$$

$W[n]$ is a zero-mean white process with variance $\sigma_W^2 = 1$.

$$Suggestion: \quad R_X[0] = 4.0917, \quad R_X[1] = 3.5273$$

1. a) Find the system function $L(z)$ and the values of the 3 poles.

   b) Find the frequency response $L(e^{j2\pi f})$ of the innovation filter.

   c) Find the PSD $S_X(e^{j2\pi f})$ of $X[n]$ and plot its progress.

Optimal 1-Step Linear Prediction: AR(3) Process

2. a) Find the 1-step optimal linear predictor of order P=1 for of $X[n]$, that is the optimal linear estimator:

$$\hat{X}[n] = h[1]X[n-1]$$

that minimizes the MSE:

$$\sigma_\varepsilon^2 = E\left\{\varepsilon^2[n]\right\} = E\left\{\left(X[n] - \hat{X}[n]\right)^2\right\}$$

b) Find the value of the MSE of such predictor.

3. Find the 1-step optimal linear predictor of $X[n]$, that is the optimal linear estimator:

$$\hat{X}[n] = \sum_{k=1}^{K} h[k]X[n-k]$$

with a suitable K, possibly also $K=+\infty$, and the value of its MSE.

K=3 (ottimo)

Optimal 1-Step Linear Prediction: AR(3) Process

- **Solution:**

1.a) Find the system function $L(z)$ and the values of the 3 poles.

1.b) Find the frequency response $L(e^{j2\pi f})$ of the innovation filter.

1.c) Find the PSD $S_X(e^{j2\pi f})$ of $X[n]$ and plot its progress.

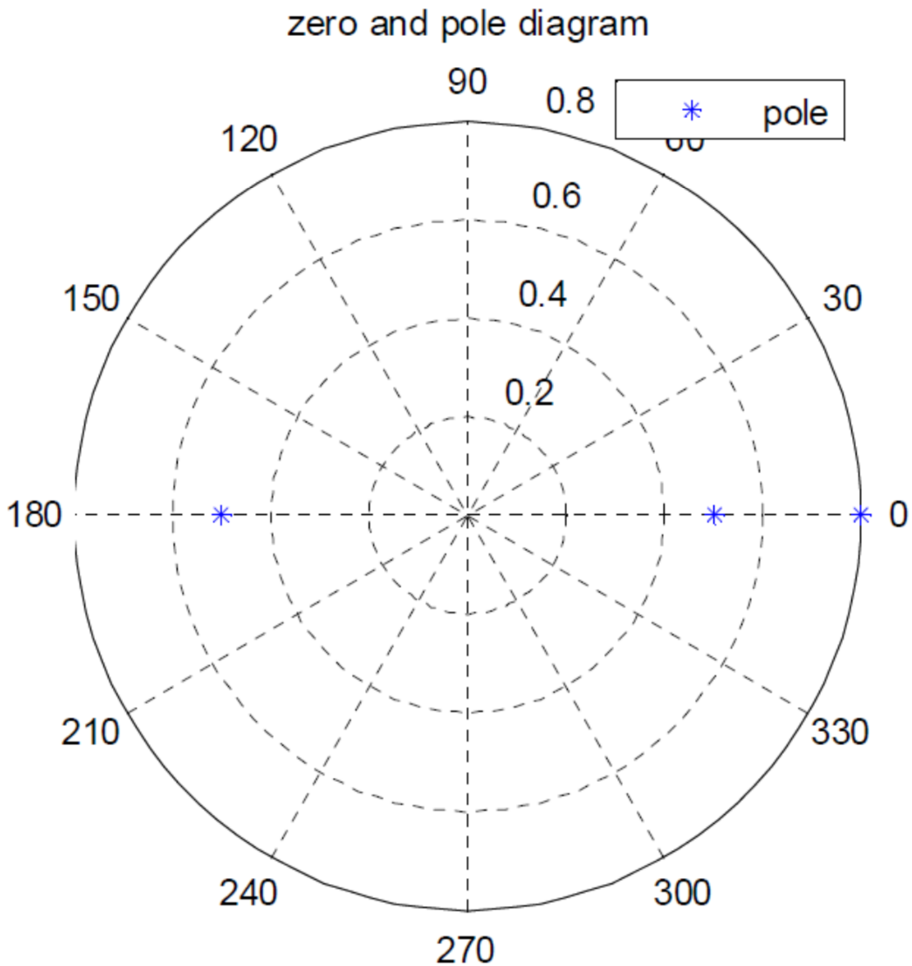
$$X[n] = \sum_{k=1}^{3} a_k X[n-k] + W[n] \quad \text{with} \quad a_1 = 4/5, a_2 = 1/4, a_3 = -1/5$$

$$X(z) = \sum_{k=1}^{3} a_k z^{-k} X(z) + W(z) \quad \rightarrow \quad L(z) = \frac{X(z)}{W(z)} = \frac{1}{1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3}} = \frac{z^3}{z^3 - a_1 z^2 - a_2 z - a_3}$$

$$L(z) = \frac{z^3}{z^3 - \frac{4}{5}z^2 - \frac{1}{4}z + \frac{1}{5}} = \frac{z^3}{z^2\left(z - \frac{4}{5}\right) - \frac{1}{4}\left(z - \frac{4}{5}\right)} = \frac{z^3}{\left(z^2 - \frac{1}{4}\right)\left(z - \frac{4}{5}\right)} = \frac{z^3}{\left(z + \frac{1}{2}\right)\left(z - \frac{1}{2}\right)\left(z - \frac{4}{5}\right)}$$

Optimal 1-Step Linear Prediction: AR(3) Process

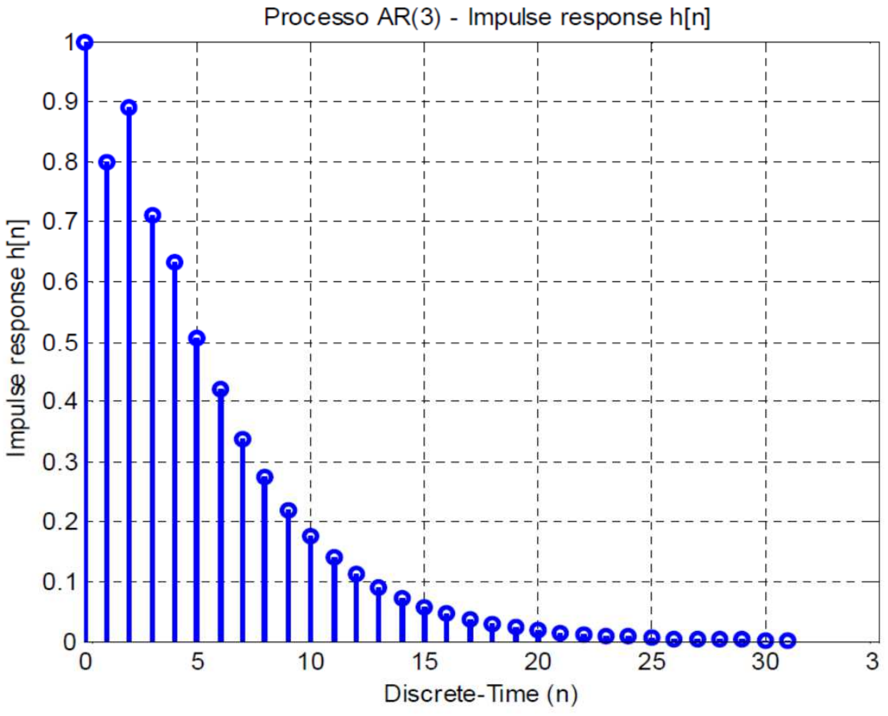
$p_{1,2} = \pm\frac{1}{2} = \pm 0.5, \quad p_3 = \frac{4}{5} = 0.8 \quad \rightarrow$ Low pass process with wide bandwidth

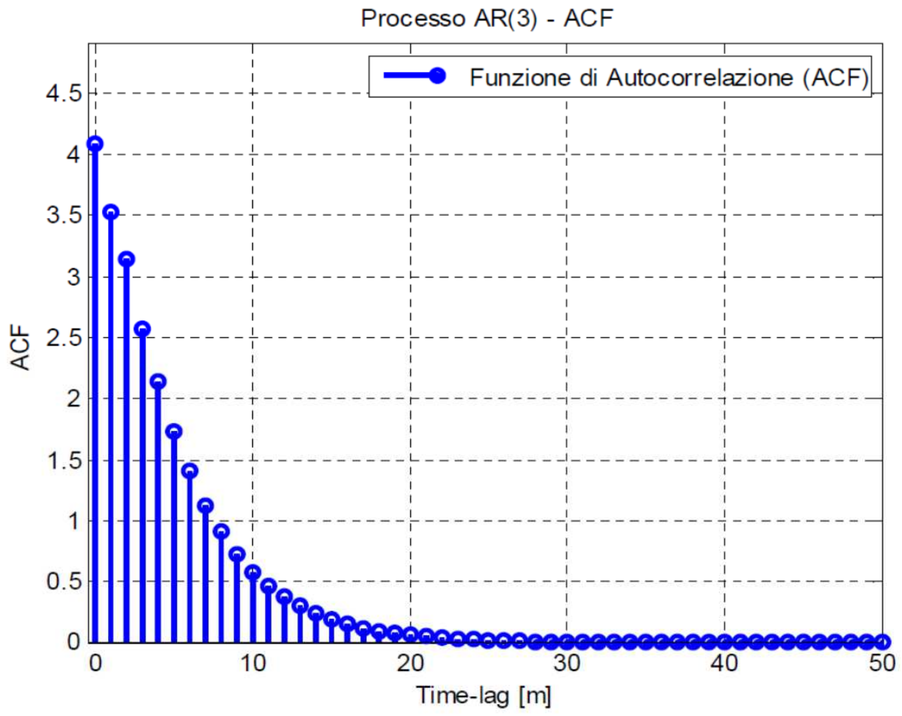


Optimal 1-Step Linear Prediction: AR(3) Process

- **Impulse response and Autocorrelation Function (ACF)**

$$R_X[0] = 4.0917, \quad R_X[1] = 3.5273$$





- **Power Spectral Density (PSD)**

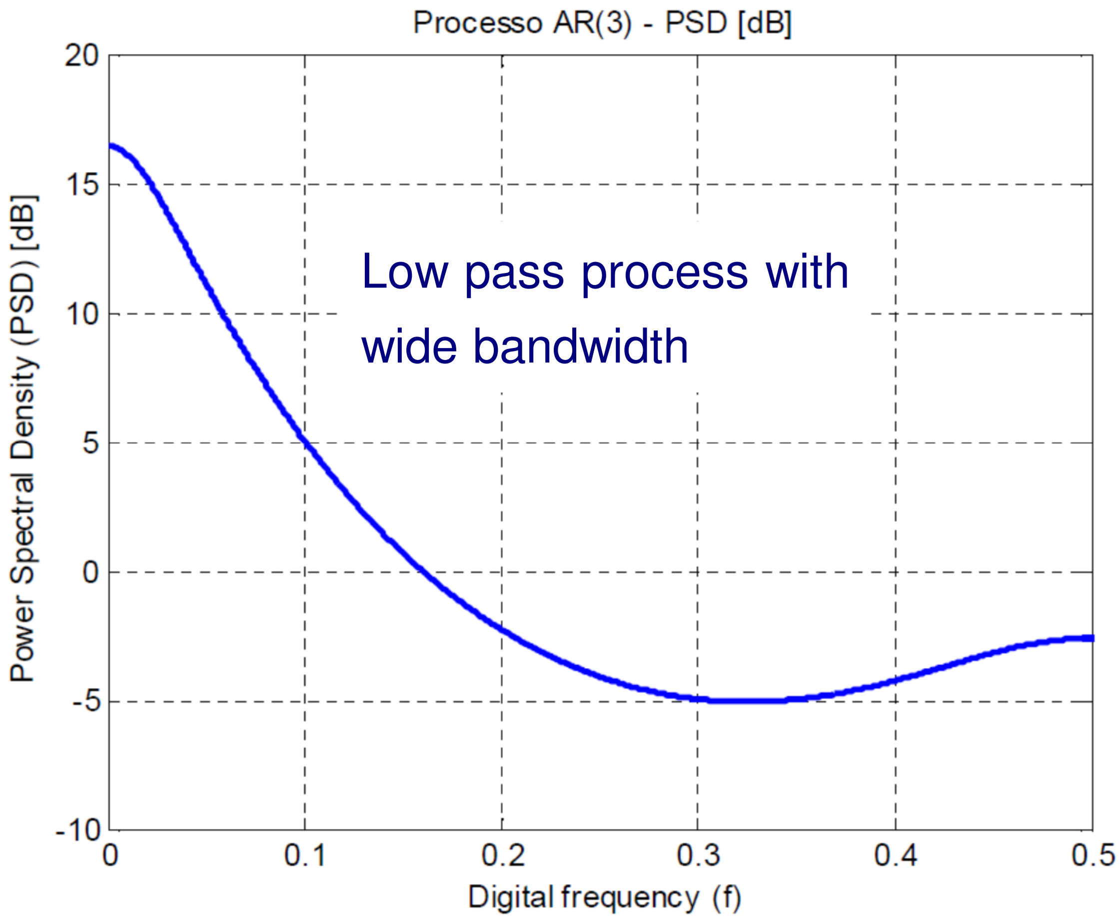
$$L(z) = \frac{z^3}{\left(z+\frac{1}{2}\right)\left(z-\frac{1}{2}\right)\left(z-\frac{4}{5}\right)} \quad \rightarrow \quad L\left(e^{j2\pi f}\right) = L(z)\Big|_{z=e^{j2\pi f}}$$

$$L\left(e^{j2\pi f}\right) = \frac{e^{j6\pi f}}{\left(e^{j2\pi f}+\frac{1}{2}\right)\left(e^{j2\pi f}-\frac{1}{2}\right)\left(e^{j2\pi f}-\frac{4}{5}\right)} \quad \rightarrow \quad \left|L\left(e^{j2\pi f}\right)\right| = \frac{1}{\left|e^{j2\pi f}+\frac{1}{2}\right|\left|e^{j2\pi f}-\frac{1}{2}\right|\left|e^{j2\pi f}-\frac{4}{5}\right|}$$

$$S_X\left(e^{j2\pi f}\right) = \sigma_W^2\left|L\left(e^{j2\pi f}\right)\right|^2 = \frac{1}{\left|e^{j2\pi f}+\frac{1}{2}\right|^2\left|e^{j2\pi f}-\frac{1}{2}\right|^2\left|e^{j2\pi f}-\frac{4}{5}\right|^2}$$

Optimal 1-Step Linear Prediction: AR(3) Process

Power Spectral Density (PSD)



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2.a) Find the 1-step optimal linear predictor of order P=1 of $X[n]$, that is the optimal linear estimator that minimizes the MSE.

2.b) Find the value of the MSE of such predictor.

$$\hat{X}[n] = h[1]X[n-1]$$

$h[1]: \quad E\left\{\varepsilon[n]X[n-1]\right\} = 0$ Orthogonality Principle

$$E\left\{\varepsilon[n]X[n-1]\right\} = E\left\{\left(X[n]-h[1]X[n-1]\right)X[n-1]\right\} = R_X[1]-h[1]R_X[0] = 0$$

$$h[1] = \frac{R_X[1]}{R_X[0]} = \rho_X[1] = \frac{3.5273}{4.0917} = 0.8621$$

$$\sigma_\varepsilon^2 = \left\{\varepsilon^2[n]\right\} = E\left\{\varepsilon[n]X[n]\right\} = E\left\{(X[n]-h[1]X[n-1])X[n]\right\} = R_X[0]-h[1]R_X[1]$$

$$= R_X[0] - \frac{R_X^2[1]}{R_X[0]} = R_X[0]\left(1-\frac{R_X^2[1]}{R_X^2[0]}\right) = R_X[0]\left(1-\rho_X^2[1]\right) = 4.0917\left(1\text{-}(0.8621)^2\right) = 1.0507$$

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3. Find the optimal 1-step linear predictor of $X[n]$, that is the optimal linear estimator and the value of its MSE.

$$\hat{X}[n] = E\left\{X[n] \middle| X[n-1], X[n-2], X[n-3], \cdots\right\}$$

$$= E\left\{\sum_{k=1}^{3} a_k X[n-k] + W[n] \middle| X[n-1], X[n-2], X[n-3], \cdots\right\}$$

$$= \sum_{k=1}^{3} a_k X[n-k] + E\left\{W[n] \middle| X[n-1], X[n-2], X[n-3], \cdots\right\}$$

$$= \sum_{k=1}^{3} a_k X[n-k] + E\left\{W[n]\right\} = \sum_{k=1}^{3} a_k X[n-k]$$

$$= \frac{4}{5} X[n-1] + \frac{1}{4} X[n-2] - \frac{1}{5} X[n-3]$$

$$\varepsilon[n] = X[n] - \hat{X}[n] = W[n] \quad \rightarrow \quad \sigma_\varepsilon^2 = \sigma_W^2 = 1$$